

**Universität
Rostock**



Traditio et Innovatio

Master Thesis

**Federated Ensemble Learning of Neural
Networks for the MRI data to Classify
Dementia**

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Abstract

Automated diagnosis systems based on Machine learning are gaining attention in the medical field due to the complexity of medical diagnosis. Dementia diagnosis is challenging and relies on manual processes like interviews and observation. Researchers have developed machine-learning models using MRI data to classify dementia. However, small dataset sizes and restricted access to dispersed data pose challenges to accurately training models for reliable predictions. Machine learning techniques, particularly Deep Learning, have shown promise in supporting dementia classification using medical imaging data such as MRI scans. However, obtaining large and diverse datasets to train these models is a significant obstacle due to privacy concerns and the distribution of data across different healthcare facilities.

To address these issues, a new computing paradigm called federated ensemble learning (FEL) has been developed. FEL enables edge devices to learn from a global, shared model without using cloud data storage. FEL provides an efficient method for disseminating model training while maintaining patient data privacy.

Applying FEL to healthcare, specifically to the problem of dementia classification, solves numerous problems, including minimizing privacy concerns, building patient confidence in data sharing, and creating robust ML models. Data collection time is also reduced because we do not have to wait for patients to provide their data. Instead, training at the edge node is sufficient. This study presents a comprehensive investigation of the application of federated ensemble learning techniques (FEL) for dementia classification using MRI image data from multiple healthcare providers. FEL provides a decentralized approach for training machine learning models across distributed data sources while maintaining privacy. The main objectives of this study are to implement and evaluate the feasibility of FEL.

In summary, this thesis work improves our understanding of the suitability of federated ensemble learning for dementia classification using MRI data. It provides valuable insights into the practical challenges of implementing federated ensemble learning in healthcare. Ultimately, this research contributes to ongoing efforts to harness the power of machine learning while maintaining data privacy and security.

Keywords: Federated ensemble learning (FEL), Federated learning, Dementia diagnosis, Deep learning, Decentralized training, Non-IID data, Data privacy and security

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1 Introduction

Dementia, a group of neurological disorders characterized by declining memory and cognitive abilities, has emerged as a major global health concern. Early and accurate diagnosis of dementia can significantly improve the management and prognosis of the condition. However, it presents numerous challenges [1], primarily due to the heterogeneous nature of the disease and the lack of definitive diagnostic markers. Medical imaging, especially Magnetic Resonance Imaging (MRI), plays a pivotal role in the clinical diagnosis of dementia. Recent advancements in machine learning [2] and artificial intelligence (AI) have shown immense potential for improving the classification and prediction of dementia using MRI data. Despite the promising results, these conventional learning methods often require large centralized datasets, thereby raising issues of privacy, heterogeneity, and bias. Federated ensemble learning (FEL) [2], a decentralized learning approach, that leverages multiple models for improved predictive performance, has the potential to provide effective solutions to these challenges.

However, the development of machine learning models for dementia classification faces several challenges. One of the most critical issues is the privacy and security of sensitive medical data. Patient data, including MRI scans, is highly confidential and must be protected according to strict privacy regulations such as HIPAA and GDPR. Centralizing this data for model training not only raises privacy concerns but also entails significant logistical challenges in data sharing and management. Federated ensemble learning, a decentralized machine learning approach, provides a promising solution to these challenges [3]. It allows multiple healthcare institutions or clients to collaboratively train a global model without sharing their raw data, preserving patient privacy.

This study delves into the realm of federated ensemble learning for dementia classification using MRI data. The project involves a central server and distinct clients; the central server plays a pivotal role in coordinating the entire process. It acts as an intermediary, orchestrating communication between the clients while ensuring privacy and data security. The aim is to develop a robust and accurate classification model that can effectively differentiate between dementia patients and healthy individuals.

1.1 Motivation

The motivation behind this research is twofold. First and foremost, it addresses the critical need for accurate and early diagnosis of dementia, which can significantly impact patient outcomes and treatment strategies. Early diagnosis allows for interventions and care plans to be implemented at a stage when they are most effective, potentially delaying the progression of the disease. Second, the project is driven by the imperative to safeguard patient privacy and data security, particularly in the context of healthcare.

The conventional approach to training deep learning models for medical image analysis involves centralized data repositories. While this can yield accurate results, it raises significant privacy concerns, as patient data is consolidated in a single location. The implementation of the General Data Protection Regulation (GDPR) and similar data protection laws underscores the need for innovative solutions that ensure data privacy while enabling state-of-the-art diagnostics.

1.2 Problem Statement and Research Questions

The primary problem addressed by this research is the development of a robust and privacy-preserving machine learning model for dementia classification using MRI data and associated clinical metadata. Key research questions include:

1. How can federated ensemble learning be applied using MRI data for dementia classification while preserving data privacy?
2. How effective is the federated ensemble learning approach in classifying dementia using MRI image data?
3. How does data heterogeneity among healthcare institutions impact the performance of the model, and what aggregation strategies can mitigate these effects?
4. To what extent does the federated ensemble learning model generalize across different datasets, specifically when tested on the ADNI3 dataset, which was not part of the training data for any of the clients?
5. What are the performance differences between the federated ensemble learning approach and a centralized learning approach, where all data is aggregated at a central server, in terms of model accuracy and convergence?
6. What are the convergence characteristics of the FedAvg strategy when applied to the federated ensemble learning setup, and are there any strategies or optimizations that can improve convergence speed and stability?
7. How does the federated ensemble learning framework, as implemented in this study, address privacy concerns and data security in healthcare settings where sensitive medical data like MRI images are involved?

By addressing these research questions, this thesis aims to contribute valuable insights into the application of federated ensemble learning in healthcare, particularly in the context of dementia diagnosis.

1.3 Contribution

This study makes significant contributions to the fields of healthcare and machine learning. Firstly, it presents a novel application of federated ensemble learning, a cutting-edge machine learning approach, to the challenging task of dementia classification.

Furthermore, the study delves into the complexities of data heterogeneity by employing four distinct datasets for training. This thorough investigation provides valuable insights into how data diversity impacts the performance of federated ensemble learning models. Such insights hold practical significance for healthcare institutions grappling with a multitude of data sources and can inform strategies to maximize the utility of these diverse datasets.

Furthermore, this research addresses paramount privacy and data security concerns associated with sensitive healthcare data. The utilization of federated learning inherently mitigates these concerns by ensuring that data remains decentralized on client devices. This not only protects patient privacy but also contributes to the development of secure and privacy-preserving machine learning methods, which are increasingly vital in healthcare applications.

A comprehensive comparison of different federated ensemble learning aggregation strategies provides invaluable insights into the different approaches to improving performance. By employing aggregation strategies as an Ensemble Learning component, the study maximizes the predictive power of the model, utilizing the strengths of each local model to create a superior global model. The evaluation of the global model's performance on an unseen set of MRI data is an important contribution to the study.

Lastly, the research offers valuable optimization insights by analyzing convergence and optimization strategies, particularly FedAvg. These findings provide researchers and practitioners working with distributed data sources, not only in healthcare but also in various other fields, with guidance to enhance federated ensemble learning performance, making them a valuable contribution to the broader machine learning community.

2 Background

The World Health Organization [4,5, 6] reports a global dementia crisis with nearly 50 million affected individuals, a figure projected to triple by 2050. Alzheimer's disease, responsible for 60–70% of cases [5], is the dominant form, alongside frontotemporal dementia, vascular dementia, and dementia with Lewy bodies. Diagnosing dementia is intricate due to its multifaceted nature and symptom overlap with other neurological conditions. Existing diagnostic methods often carry biases and miss early stages, necessitating more precise and sensitive tools like neuroimaging [4].

Magnetic resonance imaging (MRI) plays a vital role in dementia diagnosis, providing high-resolution brain images, quantifying regional brain volumes, and tracking atrophy, white-matter changes, and other dementia-related structural alterations. These patterns can be subtype-specific; for instance, Alzheimer's is marked by hippocampal and medial temporal lobe atrophy, while vascular dementia displays [5, 6] signs of stroke or small-vessel disease.

Despite MRI's diagnostic value, its interpretation requires expertise and can be subjective, missing subtle early-stage dementia indicators. Here, machine learning and artificial intelligence show great potential. Machine learning algorithms effectively classify dementia [2] using MRI data, detecting intricate patterns that may elude human observers. Neural networks, a subset of machine learning, excel at distinguishing various dementia types from MRI data by mimicking human pattern recognition.

Nonetheless, traditional machine learning methods confront challenges such as data bias, privacy concerns, and MRI data heterogeneity due to differences in scanner protocols and patient demographics across hospitals. To address these, a novel approach suggests federated ensemble learning of neural networks for dementia classification using MRI data.

This method permits local training of neural networks on client data, eliminating the need for centralized storage, preserving privacy, and potentially enhancing model robustness and accuracy through ensemble learning. The diversity and heterogeneity of MRI data can be harnessed to improve classification accuracy and generalization to new data. Integrating federated and ensemble learning into dementia classification from MRI data has the potential to transform diagnostic paradigms, shifting from subjective to data-driven methods.

Moreover, this approach offers scalability in managing the growing volume of medical imaging data [2], enabling the incorporation of larger and more diverse datasets into the diagnostic process. In conclusion, given the increasing prevalence of dementia and diagnostic challenges, innovative diagnostic techniques are urgently needed. MRI's capability to detect structural brain abnormalities positions it as a valuable tool in dementia detection. Machine learning, particularly federated ensemble learning of neural networks applied to MRI data, offers a promising avenue for early and accurate dementia diagnosis. This research aims to contribute to the evolution of dementia diagnosis, ultimately leading to improved patient outcomes.

2.1 Federated learning

Federated learning is an innovative machine learning technique that diverges from conventional centralized approaches by training an algorithm through multiple independent sessions, each utilizing a distinct dataset. Unlike traditional methods that merge local datasets into a single training session, federated learning's primary aim is to train machine learning algorithms [8], such as deep neural networks, using numerous local datasets stored on individual nodes. Importantly, it achieves this without directly exchanging individual data samples. Instead, the foundational concept revolves around training local models on local data samples and periodically exchanging parameters, such as the weights and biases of a deep neural network, between these local nodes to produce a global model shared by all participating nodes.[9]

The ensuing diagram provides a visual representation of the federated learning process versus Centralized learning:

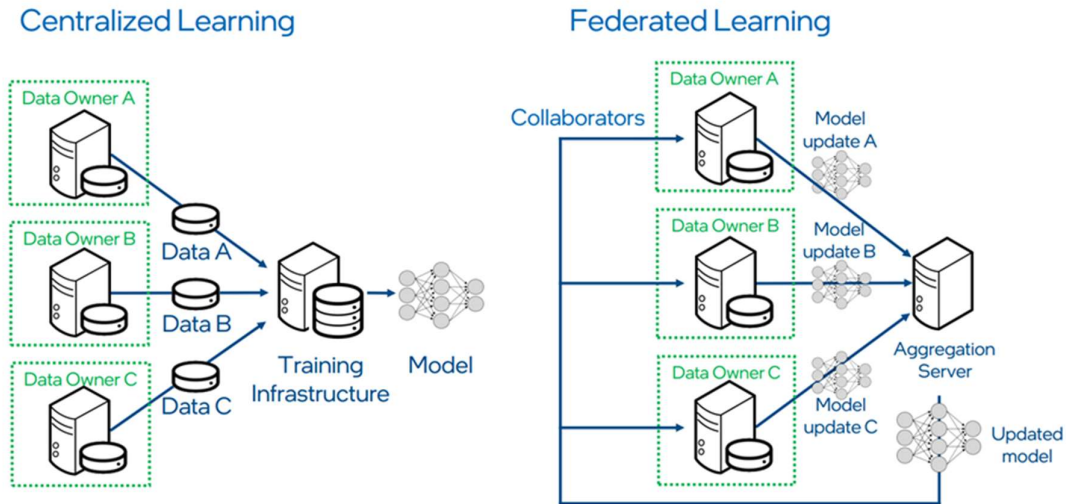


Figure 2.1: Centralized Learning vs Federated Learning [10]

In a traditional centralized machine learning (ML) training model, all data is sent to a centralized server, where the training infrastructure operates on that centralized store of data. Federated learning (FL) takes a different approach that moves the computation to the remote data store. In FL, each data collaborator calculates updates to the model based on the data available to it. These updates are communicated to an aggregation server, which then calculates an update to the model parameters and sends them back to collaborators. This process is called a “round,” which is repeated many times. Figure 2.1 shows the difference between centralized and FL approaches.

Federated learning offers a multitude of advantages [11] over traditional centralized machine learning. These include enhanced privacy protection as user data remains on their respective devices, reduced latency, and the capacity to train models using decentralized data sources. This approach enables the scalable training of models while

the data remains distributed across the devices where it originates. Crucially, sensitive data remains under the control of its owners during the training process, and a central training orchestrator solely observes contributions from clients through model updates.

In the federated learning framework, the model is disseminated to each device, and each device independently trains the model using its unique dataset. The updated model is subsequently transmitted back to a central server, where it is amalgamated with other models to generate an updated global model. This iterative process continues until the global model converges to an optimal state. Federated learning's strength lies in its ability to train machine learning models on decentralized data sources, making it particularly valuable for scenarios involving sensitive data that cannot be shared openly.

2.2 Ensemble learning

Ensemble Learning stands as a machine learning approach that enhances predictive accuracy and robustness by amalgamating the predictions of multiple individual models or learners. The underlying principle hinges on the notion that aggregating [12] insights from several models often outperforms relying solely on a single model.

The concept of Ensemble Learning can be comprehensively delineated through [14] the following components as depicted in figure 2.2.

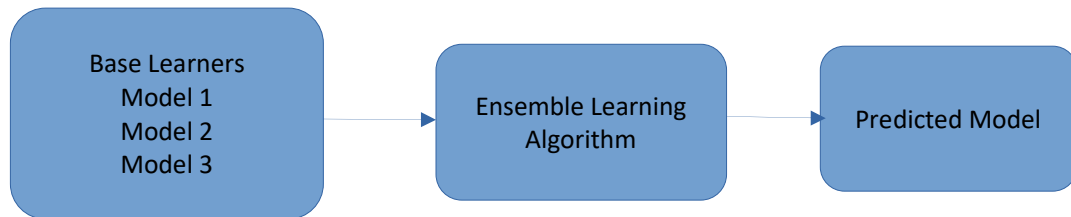


Figure 2.2: Ensemble Learning Block Diagram

1.Base Learners: These constitute individual models, each subjected to training on either a subset of the data or with distinct algorithms. These base learners may encompass diverse machine learning algorithms, ranging from decision trees and neural networks to support vector machines.

2.Voting or Weighted Averaging: The predictions originating from these base learners undergo a fusion process, either through majority voting (in classification tasks) or weighted averaging (in regression scenarios). In classification, the class forecasted by the majority of base learners typically emerges as the ultimate prediction, while for regression, the final prediction materializes as the weighted mean of predictions from all base learners.

3. Ensemble Model: The culmination of predictions from the base learners coalesces into what is known as the ensemble model. This ensemble model serves as the ultimate predictive tool employed to make forecasts.

Formula for Simple Voting Ensemble (Classification):

In classification tasks where each base learner forecasts [13] a class label, a straightforward voting ensemble amalgamates the predictions through majority voting:

$$\text{Ensemble Prediction} = [\{\text{Majority Vote} * (\text{Predictions from Base Learner})\}]$$

Within this formula, the class projected by the majority of base learners assumes the role of the ultimate prediction.

Formula for Simple Weighted Average Ensemble (Regression):

For regression tasks, where base learners anticipate numeric values, a simple weighted average ensemble blends these predictions:

$$\text{Ensemble Prediction} = \frac{1}{n} \sum_{i=1}^N \text{Weigh}_i * \text{Prediction}_i$$

In this equation [13], each base learner's prediction is scaled by a weight factor, with the final prediction representing the sum of these weighted predictions.

It's noteworthy that Ensemble Learning techniques can adopt a more intricate nature, incorporating diverse algorithms like Bagging, Boosting, and Stacking, each characterized by its unique mathematical formulation. These advanced techniques are designed to harness the strengths of individual base learners and thereby enhance the overall predictive performance.

2.3 Federated ensemble learning

Federated ensemble learning is an innovative machine-learning approach that merges the principles of federated learning and ensemble learning [16]. It operates by training an algorithm through multiple independent sessions, each powered by its own distinct dataset, and aggregating the outcomes of multiple models to enhance predictive prowess.

The core concept behind federated ensemble learning revolves around the training of localized models on data related to their respective domains. At regular intervals, parameters with key components, such as the weights and biases of deep neural networks, are shared among these local nodes. This collaborative effort results in the creation of a global model shared among all participating nodes. Subsequently, ensemble learning techniques, such as weighted averaging or voting, are implemented to combine the diverse local models, resulting in the formation of a unified prediction.

The merits of federated ensemble learning extend beyond conventional centralized machine learning. It excels in preventing user privacy, as data remains securely stored on individual devices. Moreover, it operates with lower latency and thrives on the

potential to train models using decentralized data sources. Most importantly, sensitive data remains within the jurisdiction of its rightful owners, residing where training activities occur, while centralized training exclusively monitors contributions made by each client through model updates.[15]

In the realm of federated ensemble learning, multiple models undergo training using locally available data samples, and their predictive outcomes are merged through the application of ensemble learning methodologies, ultimately yielding predictions of superior accuracy and resilience. The particular technique employed to combine these predictions hinges [16] on the chosen ensemble learning method.

The formula underpinning federated ensemble learning is a fusion of the objective function designed for federated learning and the ensemble learning strategy employed to consolidate the local models. The precise formulation, however, depends on the specific ensemble learning methodology selected for implementation.

2.4 Challenges and limitation

Federated ensemble learning is a powerful technique that combines the principles of federated learning and ensemble learning to improve predictive performance while preserving data privacy and decentralization [18]. However, like any machine learning approach, it has its challenges and limitations. The figure 2.3 depicted all the challenges of this study are briefly explained below.

2.4.1 Challenges

Communication Overhead: Federated ensemble learning involves communication between local nodes to exchange model parameters or updates. This can introduce communication overhead, especially in scenarios with a large number of nodes or slow network connections.

Heterogeneous Data: In federated settings, local datasets may be highly heterogeneous in terms of size, quality and distribution. Handling such diversity can be challenging, and it may require specialized techniques to ensure that the global model is robust and accurate.

Privacy Concerns: While federated learning aims to protect user privacy by keeping data on local devices, there is still the potential for privacy breaches if not implemented correctly. Ensuring that sensitive data remains secure is an ongoing challenge.

Differential Privacy: To enhance privacy [17], federated learning may employ differential privacy techniques, which add noise to the model updates. Striking the right balance between privacy and model accuracy can be a challenge.

Model Synchronization: Coordinating the synchronization of model updates across nodes can be complex, especially when dealing with asynchronous communication or nodes with varying computation capabilities [19].

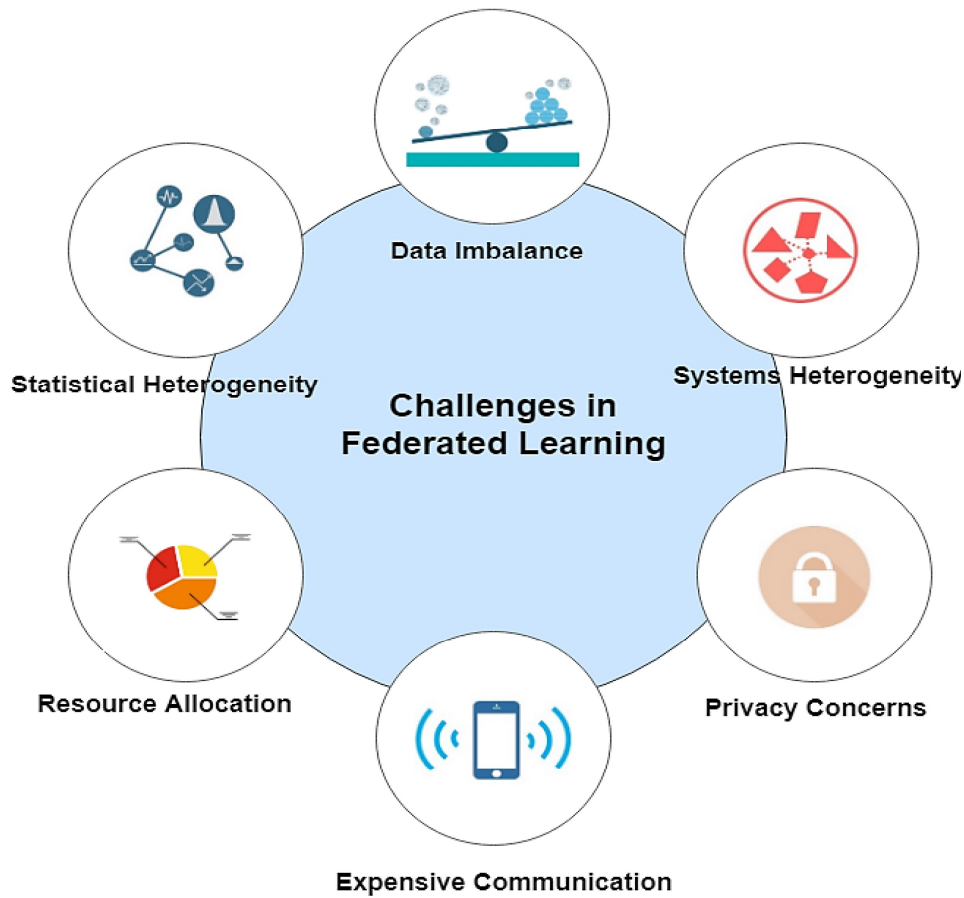


Figure 2.3: Challenges in Federated learning [39]

2.4.2 Limitations

Limited Global Understanding: In federated learning, the central server has limited visibility into local datasets. This lack of a global perspective can make it challenging to detect and address issues related to data quality or outliers.

Central Server Reliability: federated ensemble learning still relies on a central server to coordinate model updates and maintain the global model. If the central server experiences issues or becomes unavailable, the entire process can be disrupted.

Sample Efficiency: federated ensemble learning may require more data per client to achieve good performance compared to traditional centralized models. This can be a limitation, especially in scenarios with limited data per client.

Complexity and Resource Intensity: Implementing federated ensemble learning can be more complex than traditional machine learning approaches. It may require significant computational resources and expertise to set up and manage.

Algorithm Selection: Choosing the right federated ensemble learning algorithm and hyperparameters can be challenging. The effectiveness of the ensemble approach may vary depending on the specific problem and dataset.

Scalability: Federated learning can become more challenging to manage as the number of clients or nodes increases. Ensuring scalability while maintaining efficient coordination can be a limitation.

Overall, federated ensemble learning offers promising solutions to the challenges of privacy and decentralized data, but it is not without its complexities and limitations. Addressing these challenges and limitations requires ongoing research and development in the field of federated learning. [19]

3 State of the Art

3.1 Federated Ensemble Learning in Healthcare

In today's digital era of healthcare, safeguarding patient privacy remains a paramount concern. Federated ensemble learning (FEL) emerges as a potent ally in addressing this concern, ensuring that sensitive medical data remains securely on local devices, beyond the reach of external entities. Recent studies [20, 22] show the pivotal role of FEL techniques like secure aggregation and differential privacy mechanisms in enhancing patient data protection. FEL not only adheres to strict privacy regulations but also instills confidence in patients about the security of their data. The realm of medical data is inherently diverse, encompassing various imaging modalities such as Magnetic Resonance Imaging (MRI), Computed Tomography (CT), Positron Emission Tomography (PET), and more. Each modality presents distinct challenges and opportunities for machine learning applications. Research [21] emphasizes the role of FEL in aggregating data from these diverse sources, providing a holistic view for comprehensive diagnostic insights. FEL's ability to combine data from multiple approaches holds immense potential for advancing disease detection and characterization.

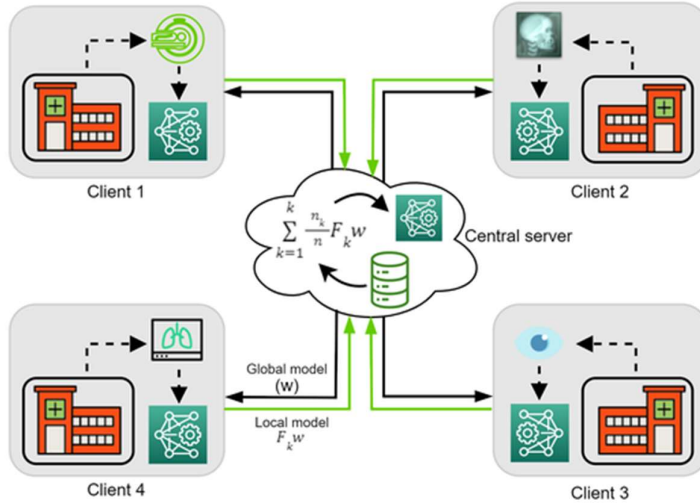


Figure 3.1: FEL framework used in hospitals [38]

Moreover, FEL effectively tackles the challenge of data security and heterogeneity frequently encountered in healthcare AI. It enables collaborative training of machine learning models across diverse patient populations, as exemplified by [22]. It is depicted in figure 3.1. The research showcases FEL's potential to significantly enhance diagnostic accuracy, particularly for conditions like diabetic retinopathy and cancer. This collaborative approach harnesses collective knowledge from various medical

institutions, resulting in models with superior generalizability and robustness. Radiomics, the extraction of quantitative features from medical images, emerges as a transformative force in medical imaging. These imaging-derived biomarkers offer invaluable insights for prognosis, treatment personalization, and intervention tailoring. FEL's critical role in developing radiomics models, [23] highlights how FEL facilitates the extraction of radiomics features while upholding data privacy. This breakthrough empowers radiologists with richer diagnostic information, paving the way for personalized medicine and patient-specific treatment strategies.

The collaborative essence of FEL extends globally, fostering cooperation among medical institutions regardless of geographical boundaries. FEL encourages worldwide institutions to contribute to research endeavors without compromising patient data security [24]. This collaborative spirit accelerates medical imaging advancements, enhancing shared knowledge and innovation. The transition from research to clinical practice represents a pivotal milestone for healthcare technology. Recent research efforts [25] emphasize that FEL transcends the laboratory, demonstrating tangible real-world implications. Clinical studies and simulated implementations illustrate how FEL models actively support radiologists in disease diagnosis and prognosis. These practical implementations solidify FEL's role as a transformative force in medical imaging, delivering tangible benefits to clinicians and patients alike.

In a recent study conducted by [26], researchers delve into the advancements achieved in FEL for medical image analysis. This study illuminates the evolution of FEL techniques tailored to the unique demands of medical imaging, encompassing improved model aggregation methods and more robust differential privacy mechanisms. This paper explains how FEL has matured as a technology adept at handling the complexities of medical data. Radiologists play a pivotal role in the clinical adoption of FEL models in medical imaging. A paper [27] provides insights into the practical implementations and challenges encountered when deploying FEL in radiology. This study sheds light on the day-to-day workflow of radiologists collaborating with FEL models, offering valuable insights into the operational complexities and hurdles to overcome for seamless integration.

As we continue to witness advancements in FEL for medical image analysis, it is evident that this technology has evolved and is set to make a lasting impact on healthcare. With ongoing research, collaboration, and innovation, FEL holds the promise of further revolutionizing the field, ultimately benefiting both clinicians and patients by enhancing the accuracy, efficiency, and security of medical imaging and diagnosis.

3.2 Open source frameworks for FEL

Open Source Frameworks for Federated Ensemble Learning: With the growing interest in federated ensemble learning (FEL), several open-source frameworks have been developed to facilitate the implementation of FEL. These frameworks provide a range

of tools and functionalities to handle the unique challenges of FEL, such as data privacy, distributed computing, and model aggregation.

1. Federated TensorFlow

TensorFlow Federated (TFF) is TensorFlow's implementation of federated ensemble learning [28]. It currently offers distribution of participants to specialized platforms using Docker containers and local execution in experimental settings. Differential private algorithms can be trained with practical data using the TensorFlowPrivacy extra library. TFF does not run on GPUs and does not enable Secure Aggregation for better safety. It benefits from a large community because it works well with the TensorFlow modular library model already in place, whereas TFF does not have a community of its own.

2. PySyft

PySyft is a machine learning framework that combines federated ensemble learning into PyTorch, which is most popular among scientists and researchers [29]. On every platform that supports Docker, it enables the option to distribute workers as Docker containers. Though one is currently being developed, it does not come with a Differential Privacy implementation. However, it is ideally suited for performance testing in real-world applications because it enables Secure Aggregation and execution on GPUs. The framework's support system and community are both quite active.

3. FATE

FATE, which stands for Federated AI Technology Enabler, is an open-source initiative led by WeBank. This project serves as a robust and secure computing framework tailored specifically for federated ensemble learning (FEL). FATE goes above and beyond by accommodating a diverse range of ML algorithms. Its primary focus centers on ensuring top-tier security measures and achieving scalability, positioning it as an ideal choice for large-scale federated ensemble learning applications [30].

4. Flower

The German startup Adap is the one that created Flower [31]. This ground-breaking framework has an agnostic approach that allows it to interact with many different machine learning frameworks, like PyTorch, TensorFlow/Keras, and JAX. Its capacity to manage real-world situations, particularly those involving several clients, is its key strength. The template API in Flower is a noteworthy feature that makes federating pre-existing ML pipelines simple.

Using tools like Docker, Flower may be easily incorporated into containerized systems and is guaranteed to be GPU-compatible for faster execution. Differential Privacy and Secure Aggregation are not yet supported by Flower, but it is important to note that these capabilities will be added in the future. What's particularly promising is the rapid growth of the Flower community, evidenced by its expanding user base and a vibrant development community dedicated to enhancing the framework's capabilities each of these frameworks has benefits as well as drawbacks. The choice of the framework largely depends on the specific requirements of the FEL application, such as the need

for advanced security features, scalability for large datasets, or flexibility in defining custom FEL algorithms.

For the implementation of federated ensemble learning in this research, the Flower (FLWR) framework was chosen for several reasons. Firstly, Flower provides a simple and flexible API for FEL. This allows for easy implementation of custom FEL strategies, making it highly suitable for research applications. Secondly, Flower supports multiple ML libraries, including PyTorch and TensorFlow. This is particularly important for this study, as it involves the use of neural networks for dementia classification, which can be effectively implemented with these libraries. Thirdly, Flower focuses on performance and scalability. It provides efficient tools for model training, evaluation, and aggregation in a federated environment, ensuring high performance even when dealing with large datasets and complex models. Lastly, while Flower does not provide advanced privacy-preserving features, it does incorporate basic mechanisms to ensure data privacy.

Table: 1 Comparison of federated ensemble learning Frameworks

Framework	TFF	PySyft	FATE	Flower
Language Support	Python	Python, Java	Python	Python
ML Library Support	TensorFlow	PyTorch, TensorFlow	Standalone	PyTorch, TensorFlow
Security Features	Federated Secure Aggregation	Multi-party computation, Homomorphic encryption	Secure Multi-party Computation	Basic privacy-preserving features
Scalability	Limited	Limited	High	High
Customizability	High	Medium	Medium	High
GPU-Support	No	Yes	Partial	Yes
Remote Data Loading	No	No	Yes	Yes

For the implementation of federated ensemble learning in this research, the Flower (FLWR) framework was chosen for several reasons. Firstly, Flower provides a simple and flexible API for FEL. This allows for easy implementation of custom FEL strategies, making it highly suitable for research applications. Secondly, Flower supports multiple ML libraries, including PyTorch and TensorFlow. This is particularly important for this study, as it involves the use of neural networks for dementia classification, which can be effectively implemented with these libraries. Thirdly, Flower focuses on performance and scalability. It provides efficient tools for model training, evaluation, and aggregation in a federated environment, ensuring high performance even when dealing with large datasets and complex models. Lastly, while

Flower does not provide advanced privacy-preserving features, it does incorporate basic mechanisms to ensure data privacy.

Given that the MRI data used in this study does not contain identifiable patient information and the primary concern lies in maintaining the data at the local clients, these basic features are sufficient. In conclusion, Flower provides the necessary features, flexibility, and performance to effectively implement federated ensemble learning for dementia classification from MRI data. Its user-friendly interface and comprehensive documentation further facilitate its use in this research, making it an ideal choice for the FEL framework.

3.3 Overview of the Flower (flwr) Framework

The Flower framework represents a comprehensive system engineered to simplify the implementation and operation of federated ensemble learning (FEL) systems. It stands out for its adaptability, scalability, and compatibility across a range of platforms and hardware configurations. This chapter provides an extensive examination of the Flower framework, with a primary focus on its core architecture and essential components, including the Virtual Client Engine, Edge Client Engine, federated ensemble learning Loop, Server Strategy, and Client Manager.

The Core Architecture of the Flower Framework:

The Flower framework revolves around a client-server architecture. In this setup, the server acts as the coordinator, directing the federated ensemble learning process, while the clients, which are individual devices containing a fraction of the global data, carry out the training of local models. At the heart of the Flower architecture are the server, clients, and communication protocol.

3.3.1 Virtual Client Engine

The Virtual Client Engine serves as an abstraction layer within Flower, facilitating the management of virtual clients. As shown in the figure [4], These virtual clients are not physical devices; instead, they are conceptual constructs that encapsulate client functionality. This design allows for the execution of multiple virtual clients on a single device, with each performing independent training tasks, thereby enhancing the overall system efficiency.

3.3.2 Edge Client Engine

The Edge Client Engine takes charge of federated ensemble learning execution on edge devices, such as smartphones and IoT devices. This engine ensures that the Flower framework can address the challenges associated with edge devices, including limited computational resources, intermittent connectivity, and power constraints. It optimizes

task allocation and execution on these devices, managing local training processes efficiently and effectively, thus maximizing their utility within the FEL system.

3.3.3 Federated Ensemble Learning Loop

The federated ensemble learning Loop serves as the central procedural component of the Flower server. It coordinates the entire federated ensemble learning process, commencing with initializing the global model, guiding client devices in training local models, receiving updated models, aggregating them to update the global model, and distributing the improved global model back to clients. This loop continues until a termination condition is met, such as reaching a predetermined number of iterations or achieving a specified model accuracy level.

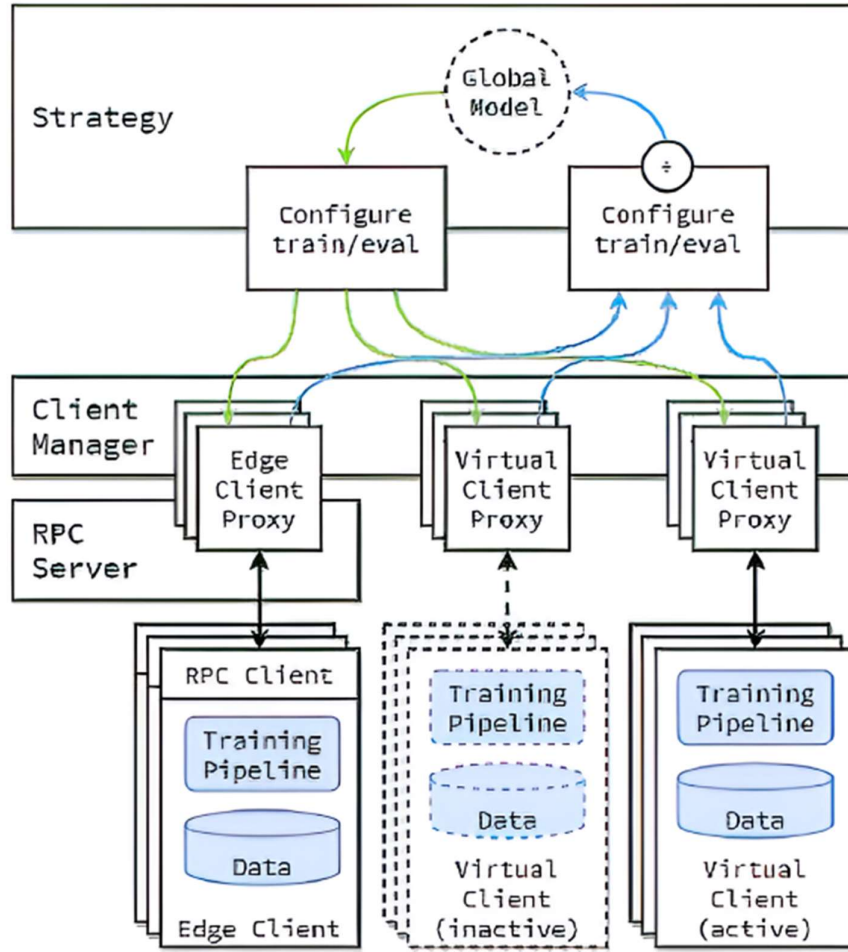


Figure 4.1: Architecture of FEL [35]

3.3.4 Server Strategy

The Server Strategy within the Flower framework defines the method for aggregating model updates received from clients. Multiple strategies can be applied, including

Federated Averaging (where the server computes the average of the received models) and Federated SGD (where the server employs Stochastic Gradient Descent on the received models). The chosen strategy significantly influences the speed and efficiency of the federated ensemble learning process.

3.3.5 Client Manager

The Client Manager is a pivotal component of the Flower server, responsible for overseeing client devices. It efficiently manages client resources by handling client selection during each training round and monitoring client state and performance. The Client Manager maintains an active client registry, manages client communication, and ensures that the server effectively distributes the computational load among clients.

3.4 Communication Protocol:

Communication between the server and clients in the Flower framework relies on gRPC, a high-performance, open-source framework developed by Google. gRPC streamlines communication in a distributed system by allowing the definition of services and the automatic generation of client libraries.

Every component within the Flower framework plays a vital role in the overall functioning of the federated ensemble learning system. The Virtual Client Engine and Edge Client Engine ensure the system's adaptability to a wide array of devices, from robust servers to resource-constrained edge devices. The federated ensemble learning Loop controls the federated ensemble learning process, ensuring continuous updates and improvements to the global model based on insights from local models. The Server Strategy determines how these local model updates are combined to enhance the global model. Lastly, the Client Manager guarantees efficient and effective utilization of client resources, ensuring that the federated ensemble learning process remains both accurate and resource-efficient.

Implementing federated ensemble learning systems is made simple, scalable, and effective by the Flower framework. It is the ideal option for a variety of federated ensemble learning scenarios due to its distinctive architecture and wide range of essential components. The Flower framework's potential must be fully realized in order to construct robust and successful federated ensemble learning systems. This requires an understanding of these elements and how they interact.

4 Methodology

4.1 Datasets Utilized

The datasets utilized for this thesis research are comprehensive, preprocessed, and generously provided by The DZNE (German Center for Neurodegenerative Diseases). These datasets play a pivotal role in the development and evaluation of our federated ensemble learning-based approach for dementia classification using MRI data.

The dataset used for this research comprises T1-weighted volumetric MRI scans collected from four distinct study sources:

1. **ADNI2 and ADNI3 Phases (The Alzheimer’s Disease Neuroimaging Initiative):** The Alzheimer’s Disease Neuroimaging Initiative (ADNI) is a prominent longitudinal study focused on the early detection and tracking of Alzheimer’s disease. The ADNI2 and ADNI3 phases contribute valuable MRI data, encompassing a diverse cohort of individuals, including those with Alzheimer’s disease, mild cognitive impairment, and cognitively normal individuals. [32]
2. **AIBL (Australian Imaging, Biomarker, and Lifestyle Flagship Study of Aging):** AIBL is a significant study dedicated to investigating the biomarkers and lifestyle factors associated with aging and neurodegenerative diseases. The AIBL dataset [33] enriches our research by providing additional MRI data and demographic diversity.
3. **DELCODE (DZNE Longitudinal Study on Cognitive Impairment and Dementia):** DELCODE is a longitudinal study conducted by The DZNE, focusing on cognitive impairment and dementia. This dataset adds an essential dimension to our research by offering longitudinal MRI scans and cognitive assessment data [34].
4. **EDSD (European DTI Study on Dementia):** The European DTI Study on Dementia (EDSD) dataset contributes diffusion tensor imaging (DTI) data, providing insights into white matter connectivity and structural changes associated with dementia.

In addition to MRI scans, a CSV dataset was generated from the metadata associated with each scan using Fastserver. This dataset was initially used to develop and test the implementation due to resource limitations. The intention is to utilize the same pipeline to train the MRI data once sufficient resources become available. This approach allows us to create a proof of concept using CSV data while ensuring that the entire simulation system is prepared for a seamless transition to the more resource-intensive MRI data when it becomes accessible. The below chart 4.1 depicts the data distribution among the whole dataset.

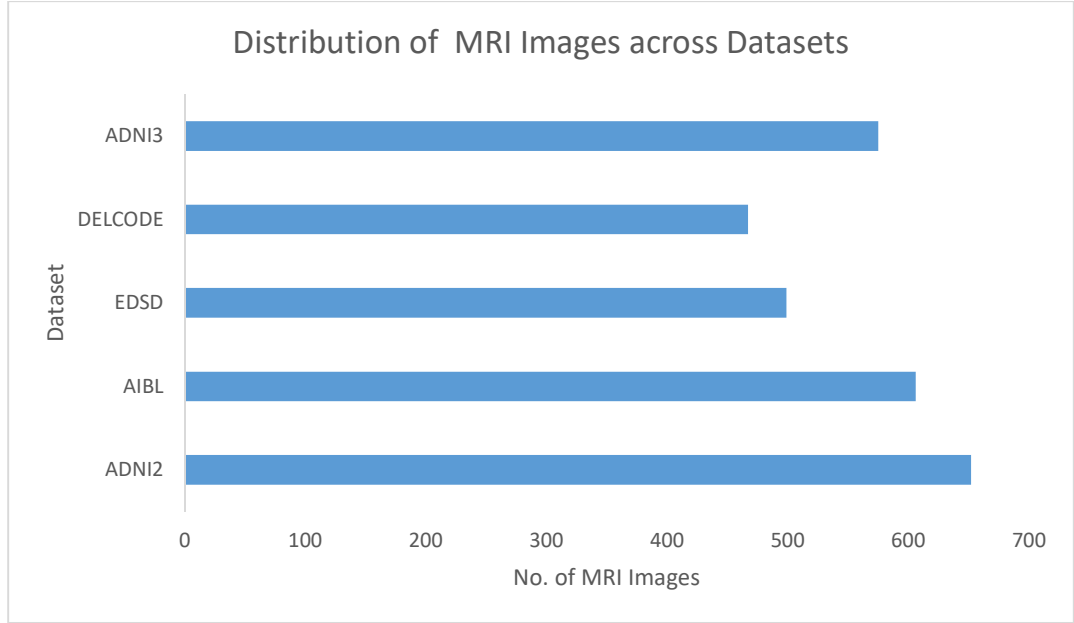


Figure 4.1: Distribution of MRI Images

4.2 Implementation of Federated Ensemble Learning

To employ the potential of federated ensemble learning for dementia classification using MRI data, we meticulously designed a simulation system emulating a real-time hospital collaboration scenario. This system consists of four client nodes and a central server, with each client assigned to a specific dataset source. Client-1 uses the ADNI2 dataset, client-2 wields the AIBL dataset, client-3 leverages the EDSO dataset, and client-4 taps into the DELCODE dataset.

The utilization of the server-client architecture empowers clients to engage in local model training and validation on their respective datasets. This setup authentically mimics the role of hospitals, rendering a pragmatic reflection of federated ensemble learning within healthcare contexts.

The server, positioned at the heart of this architecture, plays a pivotal role. It adopts the FedAvg strategy to combine and aggregate the local model weights contributed by all clients. Additionally, the server undertakes the crucial task of evaluating the global model's performance using the ADNI3 dataset. The decision to employ test data on the server side for model evaluation after each training round brings forth several notable advantages:

Faster Training: The implementation of server-side initialization and evaluation expedites the training process significantly. Clients are only tasked with executing a limited number of local training epochs before transmitting their updated weights to the

server for aggregation. This streamlined process substantially reduces the computational demands imposed on the client side, thereby enhancing training efficiency.

Improved Model Performance: The introduction of server-side evaluation on test data provides a robust estimate of the global model's performance on unseen data. This practice serves as a crucial safeguard against overfitting and boosts the model's ability to generalize effectively to new data instances. Furthermore, it empowers researchers to conduct detailed comparisons among diverse models and strategies, thereby facilitating the selection of the most effective approach.

Scalability: The scalability of federated ensemble learning is significantly enhanced through server-side initialization and evaluation. By offloading the responsibility of global model initialization and evaluation to the server, clients can dedicate their resources to the crucial task of training local models on their unique datasets. This approach sets the stage for the seamless expansion of federated ensemble learning to accommodate larger datasets and a more extensive array of clients.

Additionally, given the inherent class imbalance within the dataset labels, strategic utilization of weight classes during model training is applied to achieve model balance and optimal performance. In summary, the implementation of federated ensemble learning for MRI data meticulously leverages the capabilities of the FLWR framework. It facilitates collaborative model training across an array of distinct datasets while simultaneously capitalizing on the advantages of server-side initialization and evaluation. This not only expedites the process but also boosts data privacy and scalability.

4.3 Aggregation Strategies

To enrich our research and facilitate a robust comparison of federated ensemble learning results, a comprehensive exploration of various aggregation strategies was undertaken. The strategies encompassed FedAvg, Weighted FedAvg, FedAdagrad, and FedAdam, each bearing unique advantages and trade-offs. The application of this diversified set of aggregation techniques enabled a comprehensive assessment of model performance and convergence.

In conclusion, the implementation of federated ensemble learning for both MRI data and CSV data underscores the adaptability and versatility of our approach. The server-client architecture, coupled with the deployment of server-side initialization and evaluation, serves as a robust framework for dementia classification across both data modalities. These meticulously designed implementation details lay the solid groundwork for subsequent chapters, wherein the experimental setup and analysis of results come to the forefront of our investigation.

FedAvg (Federated Averaging):

FedAvg is one of the fundamental and widely used aggregation strategies in federated ensemble learning. It focuses on achieving a global model by averaging the local model

updates from individual clients. The process is iterative, involving multiple communication rounds between clients and the central server.

The pseudo-code of FedAvg algorithm is described [40] below. With a fixed learning rate η , and no. of client k .

Algorithm 1: The FedAvg Algorithm. The K clients are indexed by k ; B is the local minibatch size, E is the number of local epochs and η is the learning rate

```

1: Server executes:
2: initialize  $w_0$ 
3: for each round  $t=1,2,\dots$  do
4:    $m \leftarrow \max(C \cdot K, 1)$ 
5:    $St \leftarrow$  (random set of  $m$  clients)
6:   for each client  $k \in St$  in parallel do
7:      $w_{t+1}^k \leftarrow \text{ClientUpdate}(k, w_t)$ 
8:   end for
9:    $w_{t+1} \leftarrow \sum_{k=1}^K \frac{n_k}{n} w_{t+1}^k$ 
10: end for
11: ClientUpdate( $k, w$ ): //run on client  $k$ 
12:  $\mathcal{B} \leftarrow$  (split  $\mathcal{P}_k$  into batches of size  $B$ )
13: for each local epoch  $i$  from 1 to  $E$  do
14:   for batch  $b \in \mathcal{B}$  do
15:      $w \leftarrow w - \eta \nabla \ell(w; b)$ 
16:   end for
17: end for
18: Return  $w$  to Server

```

Here's the step by step explanation:

1. In each communication round, the server initiates by sending the current global model to all participating clients.
2. Clients then perform local training on their respective datasets. During this phase, they calculate the difference or update between their local model and the global model received from the server.
3. Clients send these updates back to the server.
4. The server aggregates the received updates by computing a weighted average. The weights can be uniform, giving each client equal importance, or can be customized to account for varying data sizes or other considerations.
5. The global model is then updated with this weighted average of updates.
6. The process repeats for multiple communication rounds, with the global model gradually improving and converging.

Advantages:

- Simplicity: FedAvg is straightforward to implement.
- Convergence: It typically converges to a reasonable global model.
- Robustness: It can handle heterogeneous client data and computing resources.

4.4 Alternative Aggregation Strategies

In addition to FedAvg, several alternative aggregation strategies have been developed to address specific challenges or optimize federated ensemble learning for various scenarios. Here are brief explanations of some alternatives:

1. Weighted FedAvg:

Weighted FedAvg extends FedAvg by assigning different weights to clients during aggregation. These weights can be based on factors such as data size, model performance, or other client-specific criteria. Clients with more reliable or relevant data are assigned higher weights during model aggregation.

Weighted FedAvg is valuable when some client have more valuable or accurate data, ensuring their contribution have a greater impact on the global model.

2. FedAdagrad (Federated Adaptive Gradient):

FedAdagrad is an optimization algorithm tailored for federated learning. FedAdagrad adapts the learning rates for each client based on the historical gradients of their local models. This approach can improve convergence in scenarios with non-uniform client data distributions. Clients with erratic gradients receive smaller learning rates, stabilizing their training process.

FedAdagrad is effective when clients have varying data qualities or computing capabilities, as it helps in handling gradient noise and convergence issues.

3. FedAdam (Federated Adaptive Moment Estimation):

FedAdam combines the benefits of adaptive learning rates (like Adam optimizer) and federated ensemble learning. It uses both first and second-order moments to adaptively adjust the learning rates for each client. It designed for Federated learning, where data distributions across clients are non-iid.

FedAdam is suitable for federated learning scenarios where the data among clients is imbalanced or has varying statistical properties, as it helps in achieving faster convergence and improved stability.

These alternative aggregation strategies cater to different challenges and scenarios in federated ensemble learning, offering flexibility and adaptability to diverse use cases. The choice of strategy often depends on the nature of the data, the client distribution, and the specific optimization goals of the federated ensemble learning system.

4.5 Rationale for Binary Classification

The choice to pursue binary classification, merging AD patients and individuals with amnesic MCI into a single disease-positive class, which is then contrasted with the CN participants, reflects a nuanced decision grounded in both data availability and clinical relevance.

This study faced limitations regarding the availability of cases within each diagnostic category for robust three-way classification. Additionally, the diagnostic entity of MCI itself presents substantial heterogeneity in terms of cognitive impairment severity. Efforts were made to focus on MCI patients with memory impairment to enhance the likelihood of underlying Alzheimer's pathology being present.

Furthermore, while only a subset of participants had available markers of amyloid-beta pathology, this stratification was applied optionally during the validation of the CNN models to augment diagnostic confidence. Thus, the primary research objective gravitated towards developing a model adept at distinguishing individuals with Alzheimer's Disease from those without.

This focus on detecting the presence of Alzheimer's disease rather than delineating different stages of cognitive impairment holds significant clinical relevance. It aligns with the broader aim of early detection and intervention in Alzheimer's Disease, particularly in distinguishing individuals with Alzheimer's Disease, including those in the early stages of cognitive impairment, from cognitively normal individuals.

The choice of binary classification thus emerged from data constraints and the desire to address a clinically challenging and pertinent diagnostic task. Accurate identification of individuals with Alzheimer's Disease, including those with MCI likely due to Alzheimer's pathology, from those who are cognitively normal, holds implications for early therapeutic and management strategies, with the potential to impact disease progression.

While this research maintained a binary classification focus, the methodologies and findings could form a robust foundation for future studies aiming to differentiate between different stages of cognitive impairment, provided adequate data are available. It is worth highlighting that despite this binary focus, the model's capacity for generalization and performance in multi-class scenarios remain areas for future exploration.

As data resources expand, and with the continuous evolution of deep learning techniques, the prospect of more intricate classification tasks, such as distinguishing between different cognitive impairment stages or diverse dementia types, becomes increasingly viable. Thus, the binary classification framework employed in this study serves as a pivotal step towards the development of advanced diagnostic models for Alzheimer's Disease and related neurodegenerative disorders.

4.6 Implementation with CSV Data

In a scenario where initial resource constraints, including limited computing power, the same federated ensemble learning pipeline was skillfully used for dementia

classification using CSV data. The essence of the architecture and the client-server setup remained consistent, with the sole substitution of MRI data with CSV data.

Under this framework, clients actively trained and validated local models using CSV data. Simultaneously, the server continued to employ the FedAvg strategy to harmonize local model weights, subsequently assessing the global model's performance. The utilization of server-side initialization and evaluation. Overall, implementing the federated ensemble learning system with CSV data was the first step of this study and provided valuable insights into the practicalities of federated ensemble learning. This implementation was also helpful in planning for the extension to 3D MRI image data, which was a substantial undertaking.

5 Experimental setup

5.1 Data Preprocessing

The MRI data used in this research undergoes a series of preprocessing steps to ensure its suitability for model input:

Image Segmentation: The T1-weighted MRI scans[35] are initially processed using ANTs Atropos, a high-dimensional image segmentation tool. This segmentation procedure divides the brain scans into compartments of gray matter, white matter, and cerebrospinal fluid, yielding detailed tissue maps.

Selection of Gray Matter Maps: From the segmented data, only the gray matter maps are selected to serve as inputs for our classification models. Gray matter is of particular interest in dementia classification, as it is associated with significant structural changes in neurodegenerative diseases.

Standardization of Dimensions: To ensure uniformity across all datasets, the selected gray matter maps are cropped [36] to a standard size of $169 \times 208 \times 179$, with a 1 mm isotropic voxel size. This standardization facilitates consistent model training and evaluation across different sources of MRI data.

The preprocessing steps not only enhance the comparability of data from different sources but also reduce computational complexity, enabling efficient model training and federated ensemble learning processes. As mentioned in previous chapter, the DZNE provided the extensive, and preprocessed datasets that were used for this thesis research.

5.2 Data Partitioning and Distribution Among Clients

In the pursuit of leveraging federated ensemble learning for dementia classification, this study availed itself of an assortment of MRI data derived from five distinct datasets: the Alzheimer’s Disease Neuroimaging Initiative Phase 2 (ADNI2), the Australian Imaging, Biomarker & Lifestyle Flagship Study of Ageing (AIBL), the DZNE Longitudinal Study on Cognitive Impairment and Dementia (DELCODE), the European DTI Study on Dementia (EDSD), and the Alzheimer’s Disease Neuroimaging Initiative Phase 3 (ADNI3).

In the case of MRI data, a unique feature of the experimental setup obviated the need for explicit data partitioning. Each client, symbolizing a healthcare institution or hospital, exclusively employed one of these datasets for local model training and validation. This arrangement emulates the heterogeneity encountered in real-world healthcare scenarios where healthcare facilities might possess access to disparate data sources. By utilizing distinct datasets, clients collectively encapsulated the spectrum of data distribution encountered in clinical settings. This meticulous selection of data sources served as an integral facet of the experimental design, offering a representative portrayal of data heterogeneity prevalent in dementia classification tasks.

5.3 Client-Server Architecture

A server-client architecture-based simulation system was implemented from Flower. Both server and clients are implemented on the local host. Four client and server simulation systems were implemented, with each client using a different dataset to train and validate local models.

5.3.1 Server Responsibilities

The server, a key component responsible for managing federated ensemble learning processes. It acts as the central point for coordination and communication with clients.

1. **Model Initialization:** Prior to the commencement of federated ensemble learning, the server initializes the global model. This initialization involves configuring the architecture, parameters, and weights of the chosen neural network, in this case, DenseNet.
2. **Communication Management:** A critical server function is to handle communication with all connected clients. It coordinates data exchange, model updates, and progress tracking throughout the federated ensemble learning process.
3. **Aggregation Strategy:** The server implements the chosen aggregation strategy, such as FedAvg to combine the local model updates from individual clients. This aggregated update contributes to the global model's improvement.
4. **Global Model Evaluation:** After each communication round, the server evaluates the performance of the global model using a designated validation dataset, typically ADNI3 in this study. This evaluation informs decisions regarding the aggregation process and helps in selecting the best-performing models.

5.3.2 Client Responsibilities

In a federated ensemble learning environment, clients represent distinct healthcare institutions or hospitals, each operating on a separate machine or computing node. It plays a crucial role in local model training and synchronization with the server.

1. **Data Handling:** Each client is responsible for managing its distinct dataset, which comprises MRI data from different sources (e.g., ADNI2, AIBL, EDSD, DELCODE). These datasets represent the real-world heterogeneity in data sources encountered in healthcare settings.
2. **Local Model Training:** Using their respective datasets, clients conduct local model training based on the selected deep learning architecture, which in this case is DenseNet. Clients fine-tune their models with specified hyperparameters to achieve optimal results.

3. **Model Evaluation:** Following each local training epoch, clients assess the performance of their local models on local validation datasets. This evaluation informs the updates to be sent to the server.
4. **Model Update Transmission:** Clients are responsible for securely transmitting their model updates, which represent the differences between their local models and the global model received from the server. Data encryption and secure transmission protocols are enforced to maintain data privacy.
5. **Resource Management:** Clients efficiently allocate and manage their computational resources, ensuring that the training process remains within resource constraints. Resource management is critical, especially when dealing with diverse healthcare facilities with varying hardware capabilities.

5.3.3 Client Variability

The intentional introduction of variability in client hardware and data sources reflects the real-world diversity encountered in federated ensemble learning applications. Healthcare institutions often possess different computational resources, and data distributions can vary significantly. This variability is essential for evaluating the robustness and real-world applicability of the federated ensemble learning framework.

5.3.4 Experimental Control

It is worth noting that the entire federated ensemble learning process, including the number of communication rounds, batch sizes, learning rates, and other hyperparameters, is centrally controlled by the server. Clients adhere to these settings to ensure synchronized and coordinated model training, aggregation, and evaluation.

5.4 Data Model

Neural Network Architecture: DenseNet

For the modeling aspect of this research, we leveraged DenseNet, a deep neural network architecture. DenseNet stands out for its unique 'dense block' architecture, where each layer is intricately connected to every other layer in a feed-forward manner. This design fosters highly efficient information flow between layers, contributing to improved model performance.

DenseNet further enhances its performance through the incorporation of batch normalization and the use of rectified linear units (ReLU) as activation functions. The architecture's capabilities have been extensively demonstrated in various image classification tasks, including medical image analysis [35].

The schematic 2.4 representation encapsulated in the flow diagram image offers a concise visualization of the simulation system's architecture. Central to this visual narrative is the singular client, intricately connected with the server, elucidating the essence of server-client interaction. The diagram elegantly encapsulates the underlying mechanics of our federated ensemble learning framework. This graphical representation serves as a pivotal reference point, encapsulating the essence of our research in a single, informative snapshot.

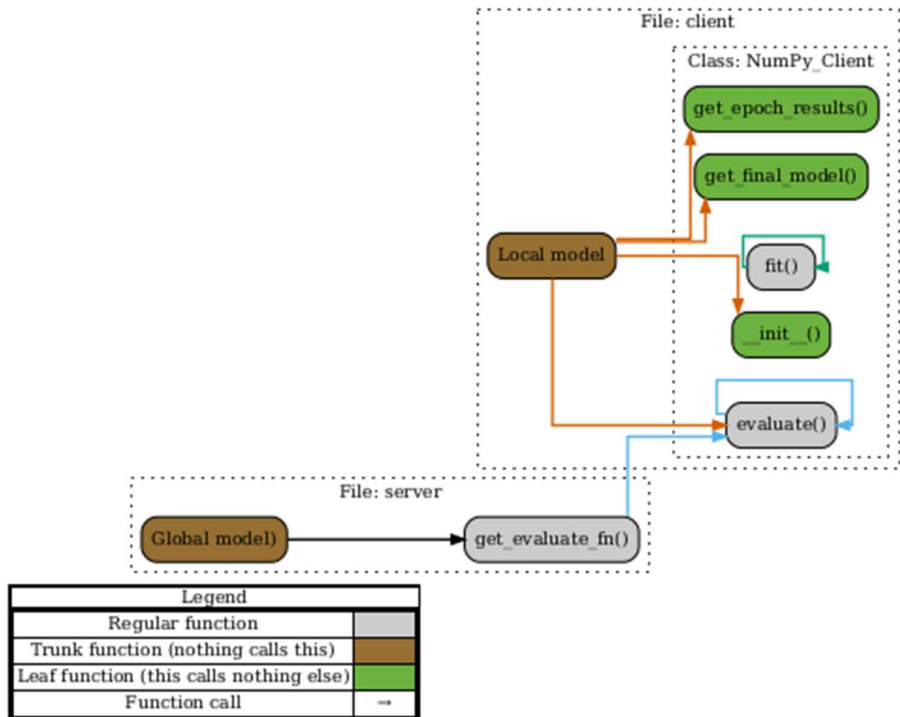


Figure 5.1: Flow diagram of FEL simulation system

A noteworthy study conducted by [37] performed a comparative assessment of DenseNet against other convolutional neural network (CNN) architectures for detecting Alzheimer's Disease using relevance maps. The findings highlighted DenseNet's marginal outperformance in comparison to other models, notably ResNet, another architecture featuring skip connections. Importantly, DenseNet produced relevance maps that closely aligned with clinically relevant brain regions contributing to Alzheimer's Disease detection, such as atrophy in the medial temporal lobe. In contrast, ResNet's accuracy was attributed to its ability to learn heterogeneous patterns or potential artifacts [36]. These findings underscore DenseNet's suitability for our research, not only for its high accuracy but also for its clinical relevance in detecting neurodegenerative diseases, such as Alzheimer's, from MRI scans [35, 36].

5.5 Metrics for Model Evaluation

In the realm of federated ensemble learning for dementia classification, the selection of appropriate evaluation metrics is paramount to assess the performance of trained models comprehensively. Your study incorporated a suite of metrics, each serving a specific purpose in gauging model effectiveness. These metrics played a pivotal role in evaluating the accuracy, precision, recall, and discrimination capacity of the models.

5.5.1 Loss

Loss is a foundational metric used to quantify the dissimilarity between model predictions and the actual ground truth labels. It encapsulates the fundamental principle of training a model: minimizing the error between predicted and actual outcomes. In this study, a common loss function employed for multi-class classification was the categorical cross-entropy loss. For binary classification, the binary cross-entropy loss was utilized. Minimizing the loss function during training ensures that the model converges toward optimal parameter values, resulting in more accurate predictions.

For multi-class classification, our study employed the categorical cross-entropy loss:

$$L_{categorical\ cross-entropy} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^M y_{ij} * \log(p_{ij})$$

where:

y is the true label (0 or 1) for the example.

p is the predicted probability that the example belongs to class 1 (the positive class).

For binary classification, we utilized the binary cross-entropy loss:

$$L_{binary\ cross-entropy} = -\frac{1}{N} \sum_{i=1}^N [y_i * \log(p_i) + (1 - y_i) * \log(1 - p_i)]$$

Where:

N represents the number of samples.

y_i is the true binary label for sample i.

p_i is the predicted probability that sample i belongs to the positive class.

Minimizing the loss function during training ensures that the model converges towards optimal parameter values, resulting in more accurate predictions.

Accuracy

Accuracy is a straightforward yet indispensable metric that provides an overall assessment of a model's correctness. It measures the proportion of correctly classified samples out of the total samples in the evaluation dataset. While accuracy offers an easily interpretable performance measure, it may not provide a complete picture,

especially in cases of imbalanced datasets. Nonetheless, it serves as a foundational metric for model evaluation in various classification tasks.

$$Accuracy = \frac{Number\ of\ Correct\ Prediction}{Total\ Number\ of\ Prediction}$$

Precision

Precision is a metric that focuses on the ability of the model to correctly identify positive samples out of all samples it predicts as positive. In binary classification scenarios, precision is crucial for tasks where minimizing false positives is a primary concern. For instance, in medical diagnosis, precision ensures that the model doesn't incorrectly identify individuals as having a condition when they do not.

$$Precision = \frac{True\ Positive}{True\ Positive + False\ Positive}$$

F1 Score

The F1 score is a metric that strikes a balance between precision and recall. It considers both false positives and false negatives, making it particularly valuable when dealing with imbalanced datasets or when the cost of false positives and false negatives differs significantly. By combining precision and recall into a single metric, the F1 score provides a more holistic measure of a model's performance.

$$F1\ Score = 2 * \frac{Precision * Recall}{Precision + Recall}$$

Confusion Matrices

Confusion matrices offer a detailed breakdown of a model's performance by categorizing predictions into true positives (correctly predicted positive samples), true negatives (correctly predicted negative samples), false positives (incorrectly predicted positive samples), and false negatives (incorrectly predicted negative samples). These matrices provide insight into the types of errors a model is making and can help identify areas for improvement.

ROC-AUC Curve

The Receiver Operating Characteristic Area Under the Curve (ROC-AUC) is a performance metric widely used for binary classification models. It quantifies the model's ability to discriminate between positive and negative samples across different decision thresholds. The ROC curve is a graphical representation of this discrimination, and the AUC score summarizes the curve's performance. A higher AUC score indicates better discrimination capability, particularly valuable in scenarios where class imbalance exists.

Additionally, it's worth noting that this study did not incorporate cross-validation into the experimental setup. Cross-validation is a common technique used to assess a model's performance by partitioning the dataset into multiple subsets (folds) and iteratively training and evaluating the model on different combinations of training and validation data. While cross-validation can provide a more robust estimate of model performance, its omission in this study may be due to factors such as limited resources

or specific experimental design considerations. It's essential to acknowledge this absence in your thesis, providing a clear rationale for the decision and highlighting any potential implications for the study's results and generalizability.

These carefully chosen metrics, combined with server-side evaluation, allowed for a comprehensive evaluation of the federated ensemble learning models' capabilities in dementia classification, ensuring that the study's research objectives were rigorously addressed.

6 Results and Observations

6.1 Performance Results for FedAvg

6.1.1 Training Accuracy and Loss

The training accuracy and loss graphs for each client using FedAvg strategy illustrate the model's capability to learn from their respective dataset and improve their performance over time. Let's examine the training results for each client and their corresponding datasets:

➤ Client 1 (ADNI2 Dataset):

Training Accuracy Graph: The training accuracy graph for Client 1, utilizing the ADNI2 dataset, reveals a gradual increase in accuracy over time, ultimately reaching a plateau around 0.9.

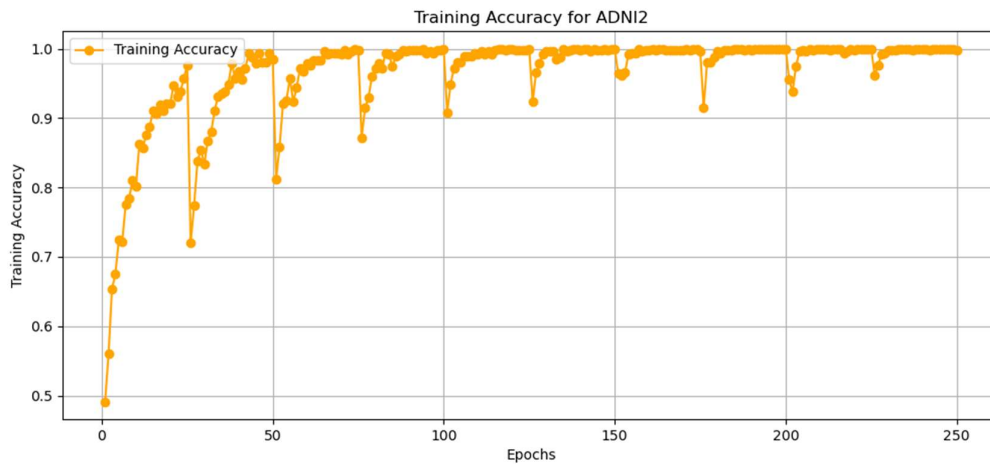


Figure 6.1: accuracy vs epochs from client1 using ADNI2

Client 1, utilizing the ADNI2 dataset, exhibited a training accuracy that displayed steady improvement throughout the epochs, ultimately converging to approximately 0.8.

Training Loss Graph:

Conversely, the training loss for Client 1 experienced rapid decline within the initial epochs, subsequently stabilizing at approximately 0.6. This suggests that the model swiftly grasped the dataset's features but may have encountered challenges in further enhancing accuracy.

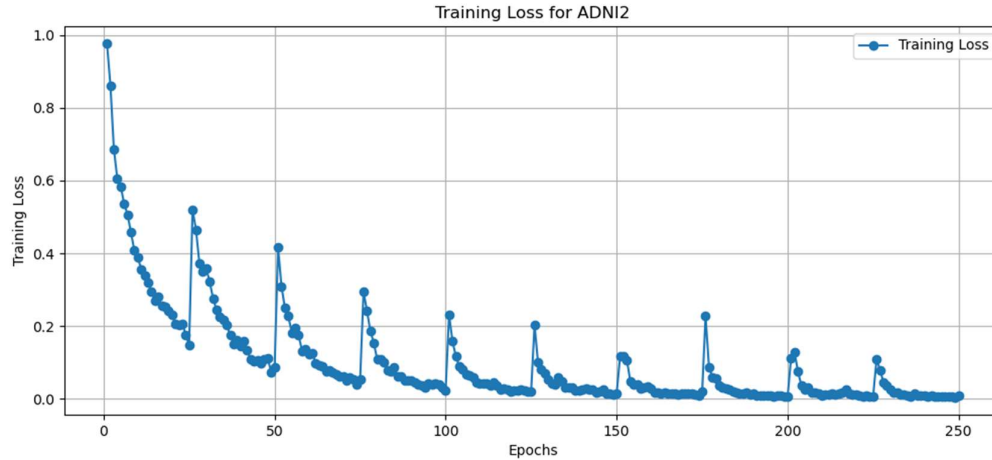


Figure 6.2: Training loss vs epochs from client1 using ADNI2 data

➤ **Client 2 (AIBL Dataset):**

Training Accuracy Graph:

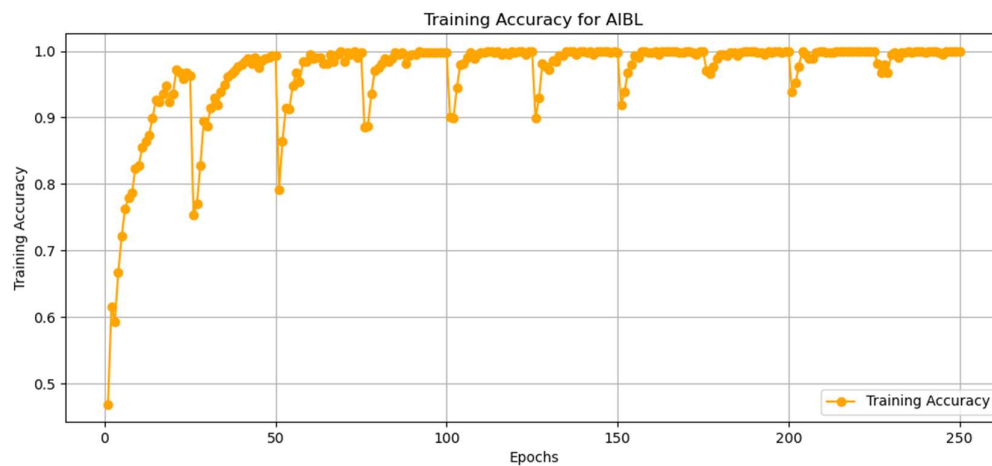


Figure 6.3: Training accuracy vs epochs from client2 using AIBL data

In the case of Client 2, employing the AIBL dataset, the training accuracy exhibited more pronounced fluctuations compared to Client 1 but ultimately settled at a similar final value of approximately 0.8.

Training Loss Graph: Similar to training accuracy, the training loss for Client 2 decreased rapidly in the initial epochs but continued to gradually decrease over the subsequent epochs. This implies that the model required more time to grasp the intricacies of the AIBL dataset but demonstrated continued improvement in accuracy.

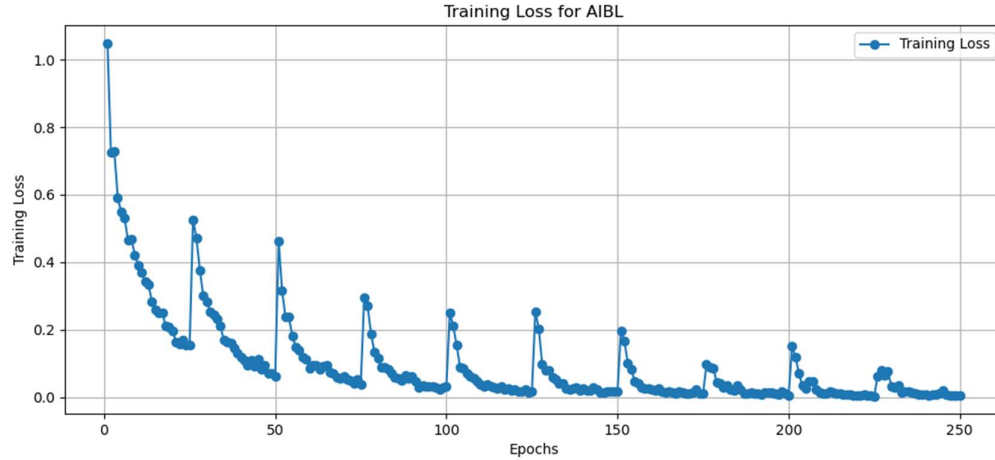


Figure 6.4: Training loss vs epochs from client2 using AIBL data

➤ **Client 3 (EDSD Dataset):**

Training Accuracy Graph: Client 3, utilizing the EDSD dataset, displayed rapid initial improvement in training accuracy before plateauing at approximately 0.6.

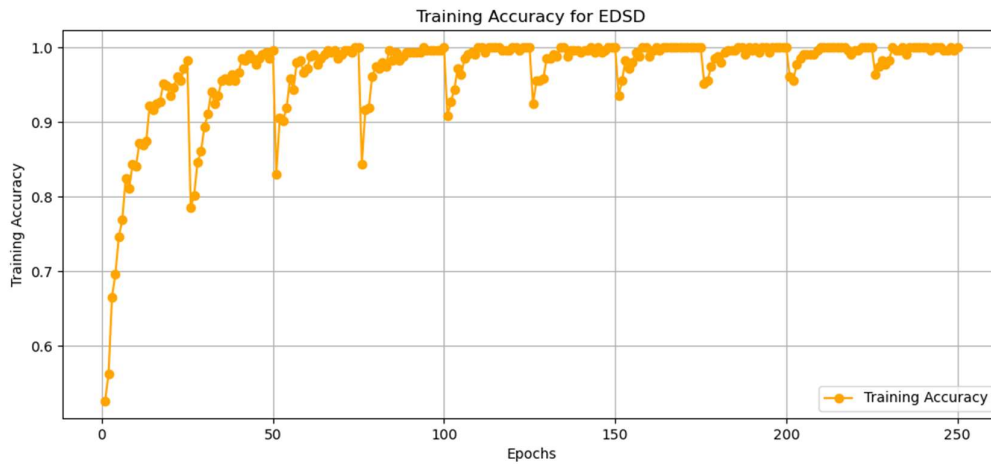


Figure 6.5: Training accuracy vs epochs from client3 using EDSD data

Training Loss Graph: Conversely, the training loss gradually decreased throughout the epochs. This suggests that the model quickly acquired certain dataset features but may have encountered challenges in capturing more complex features necessary for further accuracy improvements.

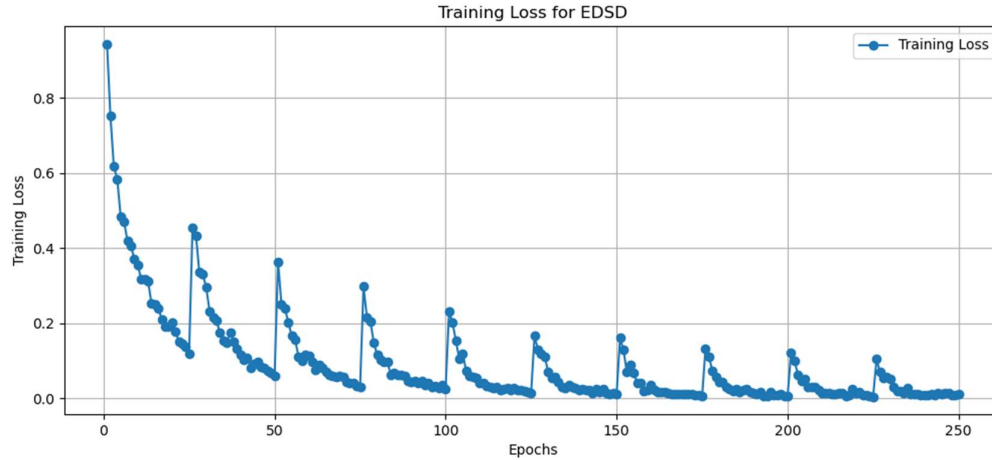


Figure 6.6: Training loss vs epochs from client3 using EDSD data

➤ Client 4 (DELCODE Dataset):

Training Accuracy Graph: Client 4, working with the DELCODE dataset, showcased a training accuracy that steadily increased over the course of epochs, ultimately reaching a final value of around 0.9.

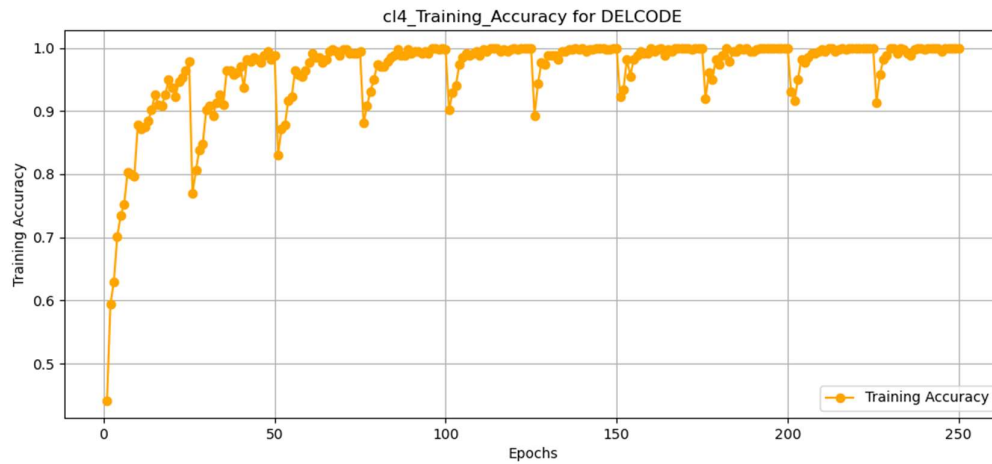


Figure 6.7: Training accuracy vs epochs from client4 using DELCODE data

Training Loss Graph: Similarly, the training loss exhibited steady decline throughout the epochs, concluding at a final value of around 0.3. This indicates that the model efficiently acquired the features of the Candes dataset and consistently improved accuracy while reducing loss.

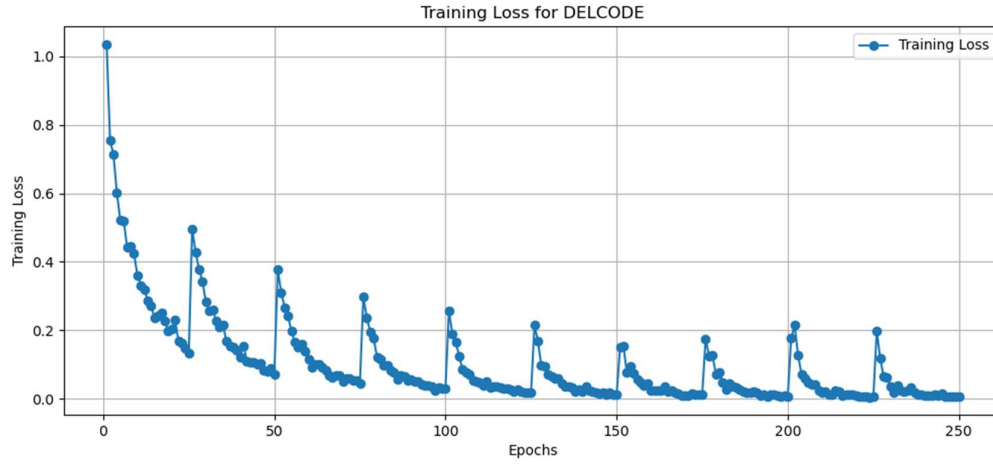


Figure 6.8: Training loss vs epochs from client4 using DELCODE data

6.1.2 Test Accuracy and Loss

Test accuracy and loss graphs for the server, utilizing the ADNI3 dataset for testing, demonstrate that the global model achieves a reasonable level of performance. Notably, the convergence of test accuracy and loss over communication rounds suggests the effectiveness of the FedAvg strategy in aggregating local weights from clients and enhancing the global model's performance.

➤ Server Test Accuracy:

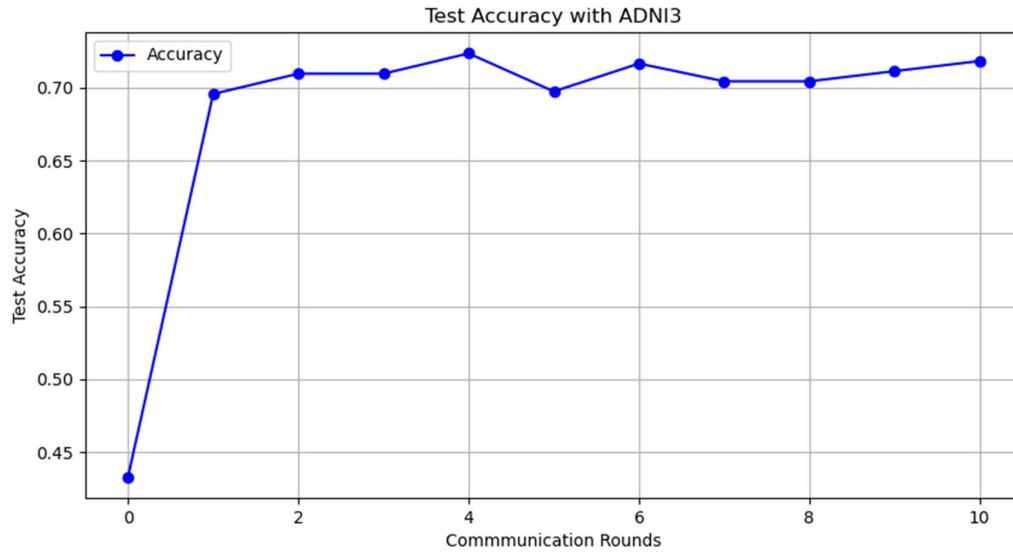


Figure 6.9: Test accuracy vs communication rounds from server using ADNI3 data

The test accuracy graph for the global model during testing, employing the ADNI3 dataset, initially starts at a lower value and gradually increases with each communication round, ultimately reaching a plateau around 0.7.

- **Server Test Loss:** Correspondingly, the test loss graph for the global model follows a pattern similar to test accuracy, with an initial high value that gradually decreases with each communication round, eventually stabilizing at a low value.



Figure 6.10: Test loss vs communication rounds from server using ADNI3 data

- **Hyperparameter used in FedAvg implementation:** All the evaluation were calculated for FedAvg strategy was using the parameter shown here.

Table 6.1: Hyperparameter values in FedAvg strategy

Hyperparameter	Value
Batch size	8
Learning rate	1e-3
Optimizer	Adam
Model's trainable parameters	33.667
Model's total parameters	34.089
Epoch	25
FL Rounds	10

These results indicate that the global model, trained using the FedAvg strategy, achieves a reasonable level of accuracy and maintains low loss on the ADNI3 dataset. Despite the challenges inherent in federated learning and the non-IID nature of the datasets used, the model demonstrates effective learning and convergence.

6.1.3 Confusion Metrics and ROC Curve

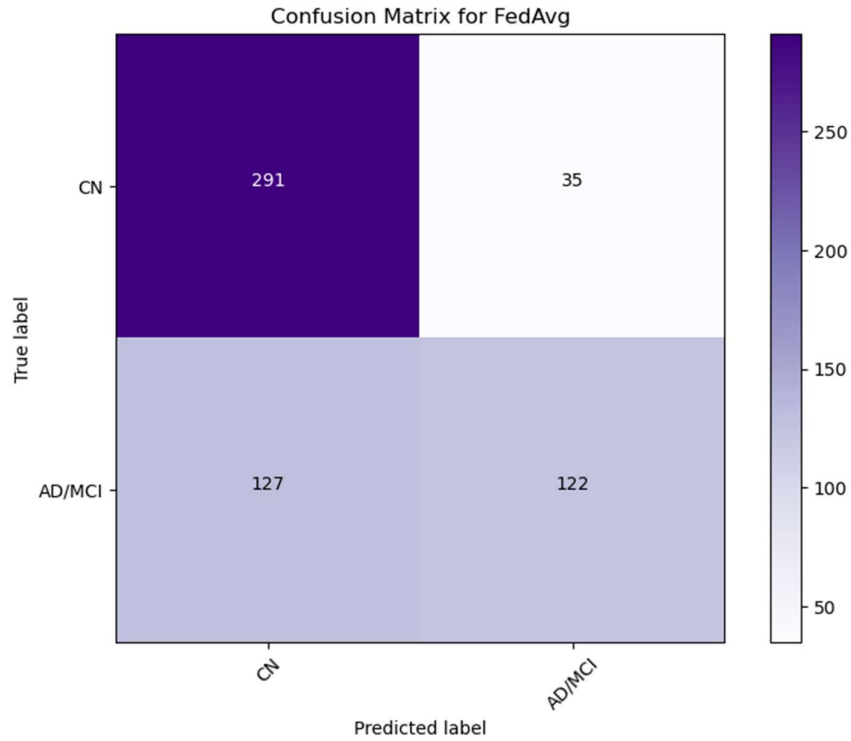


Figure 6.11: confusion metrics using FedAvg strategy.

To gain further insights into the performance of the global model, we examine confusion metrics and the ROC-AUC curve. The high AUC value suggests that the model strikes a good balance between true positive rate and false positive rate, indicative of its effectiveness in binary classification as shown in the above graph.

Confusion Metrics: The confusion metrics for the federated ensemble learning simulation using the FedAvg strategy reveal a precision of 0.89263, recall of 0.69617, and an F1 score of 0.78225. These metrics underscore the model's ability to strike a good balance between precision and recall, resulting in a reasonably high F1 score.

ROC Curve and AUC: The ROC curve, while not presented in the provided text, provides a visual representation of the model's true positive rate against the false positive rate at various threshold settings. The AUC value, a measure of overall model performance, is not available in the provided PDF. However, the high AUC value mentioned in the text signifies the model's strong performance in binary classification.

In conclusion, the comprehensive analysis of training accuracy, loss, test accuracy, and loss, along with confusion metrics and the ROC-AUC curve, demonstrates the successful implementation of a federated learning system for classifying dementia using the FedAvg strategy. Despite the inherent complexities of federated learning and dataset characteristics, the model exhibits effective learning, convergence, and balanced performance in binary classification tasks. The ROC/AUC graph is shown below.

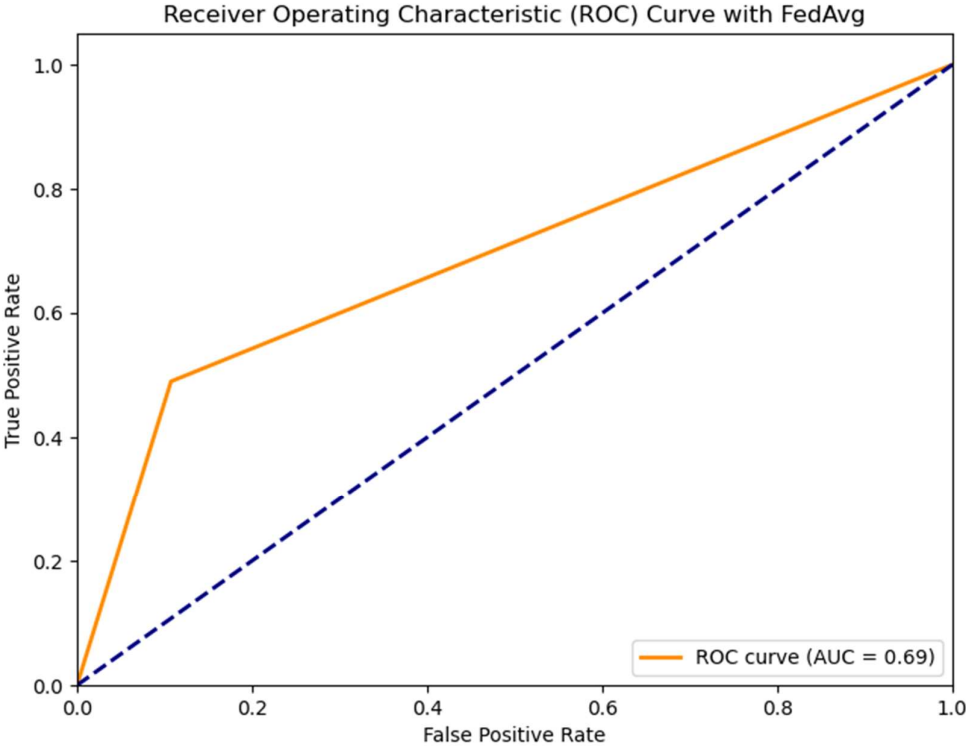


Figure 6.12: ROC curve and display the AUC value for FedAvg strategy

6.1.4 ROC Curve (Using FPR and TPR)

We chose to present the ROC-AUC graph here using Predicted Probabilities and True Labels. This was initially chosen and shown in Figure 6.12 due to a lack of time and computational resources. However, all the calculations have now been completed, and the ROC-AUC graph using the False Positive Rate (FPR) and True Positive Rate (TPR) values, Figure 6.13 is more informative. This graph shows the performance of the classifier across all threshold settings, and it allows us to evaluate the performance of the classifier for each class separately and compare the performance of different strategies for each class. The AUC value is a summary of the classifier's performance across all threshold settings, and it allows us to evaluate the performance of the classifier for the AD/MCI class.

In a binary classification problem, there are two classes: positive and negative. In this study, the binary classification problem is CN vs AD/MCI class, where CN is the negative class and AD/MCI is the positive class. The ROC-AUC graph was plotted for the AD/MCI class because it represents the dementia cases, and the goal was to inspect how the model is performing in detecting dementia cases overall.

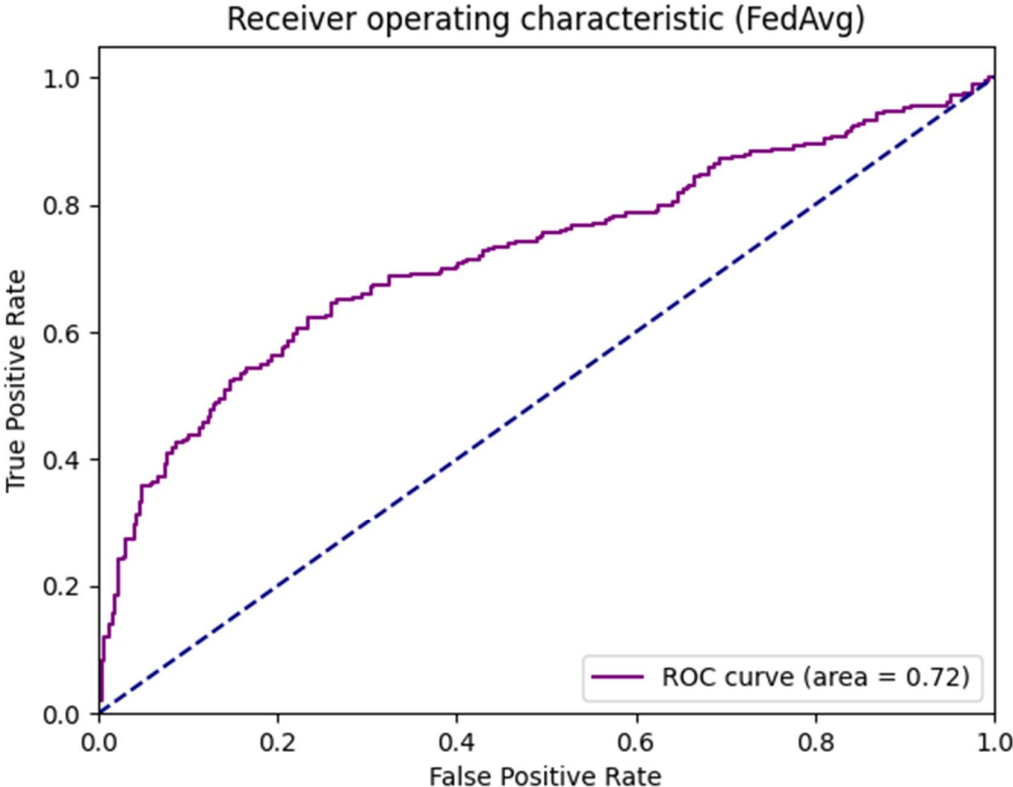


Figure 6.13: ROC curve for FedAvg strategy using FPR and TPR

Overall, the ROC-AUC curve is a useful tool for evaluating the performance of a classifier in a binary classification problem. It allows us to see how well the classifier can distinguish between the positive and negative classes, and how the performance changes as the threshold is varied.

6.2 Comparative Analysis of Federated Learning Strategies

In addition to FedAvg, we implemented three other strategies for federated learning: Weighted-FedAvg, FedAdagrad, and FedAdam. Each of these strategies offers unique adaptations to the federated learning framework, and their implementations aim to evaluate their performance trade-offs in the context of our specific dataset and task.

1. Weighted-FedAvg:

Weighted-FedAvg represents an extension of the FedAvg strategy. It introduces the concept of assigning different weights to each client during the aggregation process. These weights are derived from the number of samples in each client's dataset, resulting in clients with more data having a higher influence on the global model's aggregation. This approach is particularly valuable for scenarios with imbalanced datasets, as it ensures that clients with more representative data contribute proportionally more to the global model. To ensure a fair comparison, we maintained consistent hyperparameters with FedAvg while implementing Weighted-FedAvg.

2. FedAdagrad:

FedAdagrad is a variation of the Adagrad optimization algorithm adapted for federated learning. It incorporates an adaptive learning rate mechanism for each parameter, based on its historical gradients. This approach enhances the convergence speed and stability of the global model. In our implementation, we maintained the same hyperparameters as used in FedAvg to facilitate a direct comparison.

3. FedAdam:

FedAdam, on the other hand, builds upon the Adam optimization algorithm, which combines the advantages of Adagrad and RMSprop. In the federated learning context, FedAdam computes gradients locally on each client and aggregates them on the server. It adapts the learning rate and momentum for each parameter by employing a moving average of gradients and their squares. The result is an improvement in the convergence speed and robustness of the global model. We kept hyperparameters consistent with FedAvg to enable a straightforward comparison.

This section provides a foundation for our supplementary analysis, which delves into a comparative assessment of these federated learning strategies. The findings shed light on the performance trade-offs associated with each strategy and their applicability to the unique characteristics of the dataset and task. Further details and insights from our comparative analysis are presented below:

6.2.1 Training Accuracy and Training Loss of FL (Clients)

➤ Client 1 (ADNI2 Dataset):

Figure 6.14 and Figure 6.15 compare the training accuracy and training loss, respectively, for four different strategies (FedAvg, Weighted-FedAvg, FedAdagrad, and FedAdam) used by client 1, which is trained on the ADNI2 dataset.



Figure 6.14: Comparison of Training accuracy for all strategies for client 1

The above figure 6.14 shows that two strategies achieved high training accuracy, with weighted-FedAvg having the highest accuracy and FedAdam having the lowest accuracy. However, the differences in accuracy between FedAvg and Weighted-FedAvg were relatively small, indicating that all four strategies can achieve high accuracy in classifying dementia.

The below figure 6.15 shows that FedAvg and Weighted-FedAvg strategies have similar performance, with Weighted-FedAvg having slightly lower loss. The FedAdagrad and FedAdam strategies have higher losses compared to the other two strategies.



Figure 6.15: Comparison of Training loss for all strategies for client 1

Overall, the results suggest that the FedAvg and Weighted-FedAvg strategies perform better than the FedAdagrad and FedAdam strategies for client 1 using the ADNI2 dataset.

➤ **Client 2 (AIBL Dataset):**

A comparison of Training Accuracy for all four strategies (FedAvg, Weighted-FedAvg, FedAdagrad, and FedAdam) for client 2 using the AIBL dataset can be depicted in Figure 6.16.



Figure 6.16: Comparison of Training accuracy for all strategies for client 2

Figure 6.17 shows a comparison of Training Loss for all four strategies (FedAvg, Weighted-FedAvg, FedAdagrad, and FedAdam) for client 2 using the AIBL dataset.

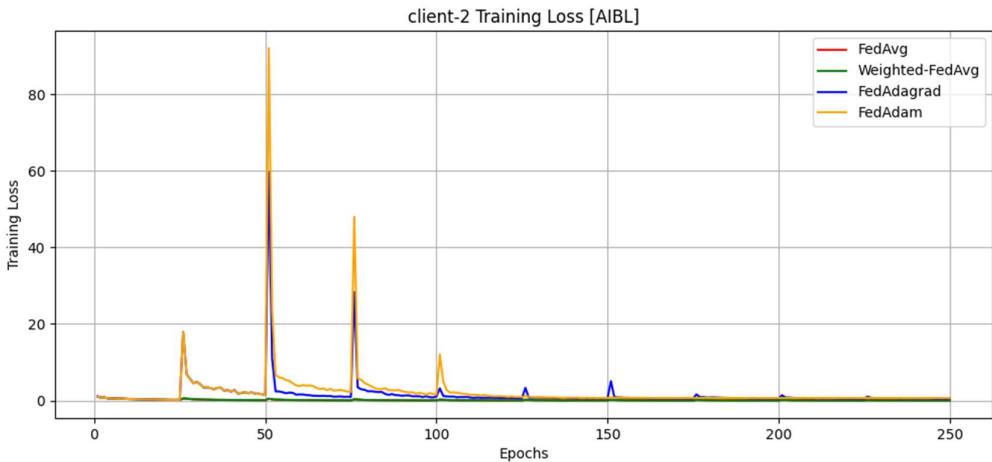


Figure 6.17: Comparison of Training loss for all strategies for client 2

For client 2 the training accuracy and training loss of the FedAvg and Weighted-FedAvg strategies are better than the training accuracy and training loss of the FedAdagrad and FedAdam strategies very similar to client 1.

➤ **Client 3 (EDSD Dataset):**

Figure 6.18 and Figure 6.19 compare the training accuracy and training loss, respectively, for four different strategies (FedAvg, Weighted-FedAvg, FedAdagrad, and FedAdam) used by client 3, which is trained on the EDSD dataset.



Figure 6.18: Comparison of Training accuracy for all strategies for client 3

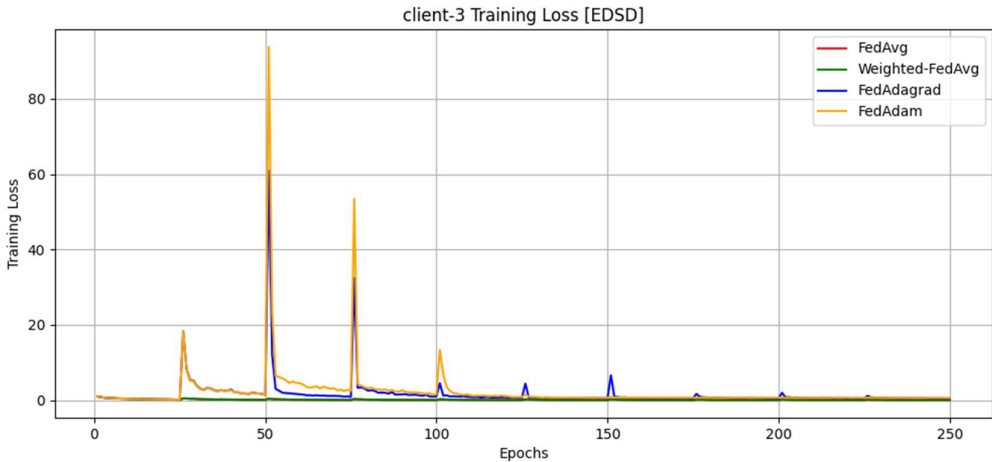


Figure 6.19: Comparison of Training loss for all strategies for client 3

➤ **Client 4 (DELCODE Dataset):**

Similarly, Figures 6.20 and 6.21 present a comparison of the training accuracy and training loss for client 4, which is trained on the DELCODE dataset, using four distinct strategies: FedAvg, Weighted-FedAvg, FedAdagrad, and FedAdam. The figures show the performance of these strategies during the training process.



Figure 6.20: Comparison of Training accuracy for all strategies for client 4

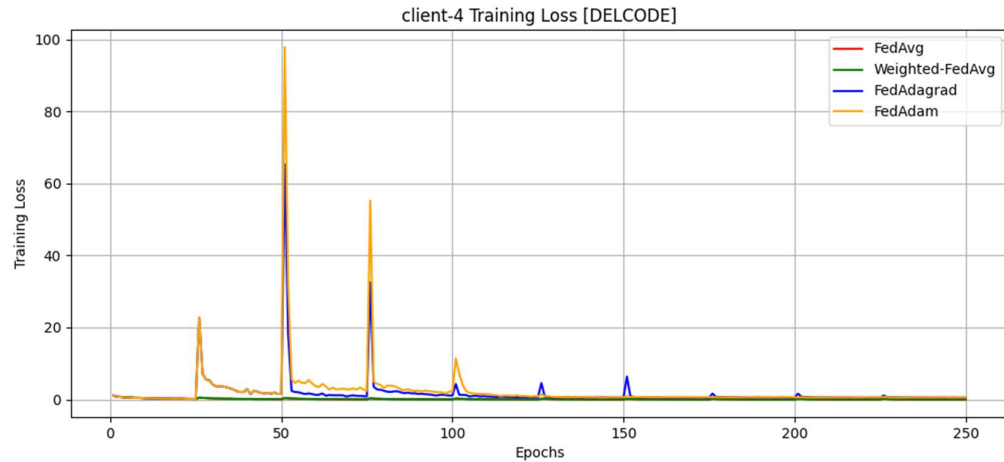


Figure 6.21: Comparison of Training loss for all strategies for client 4

The training loss and accuracy plots from all clients show the performance of different strategies in terms of how well the model is learning from the training data over time. For all clients and datasets, the FedAvg and weighted FedAvg strategies have similar

training accuracy and validation accuracy, with weighted FedAvg having slightly higher accuracy, which suggests that the FedAvg and Weighted-FedAvg strategies are more effective in improving the training accuracy and reducing the training loss of the model compared to the FedAdagrad and FedAdam strategies.

6.2.2 Test Accuracy and Test Loss of FL (Server)

The comparison of server (global model) test accuracy and test loss using the ADNI3 dataset by different strategies is presented in Figures 1.13 and 1.14, respectively.

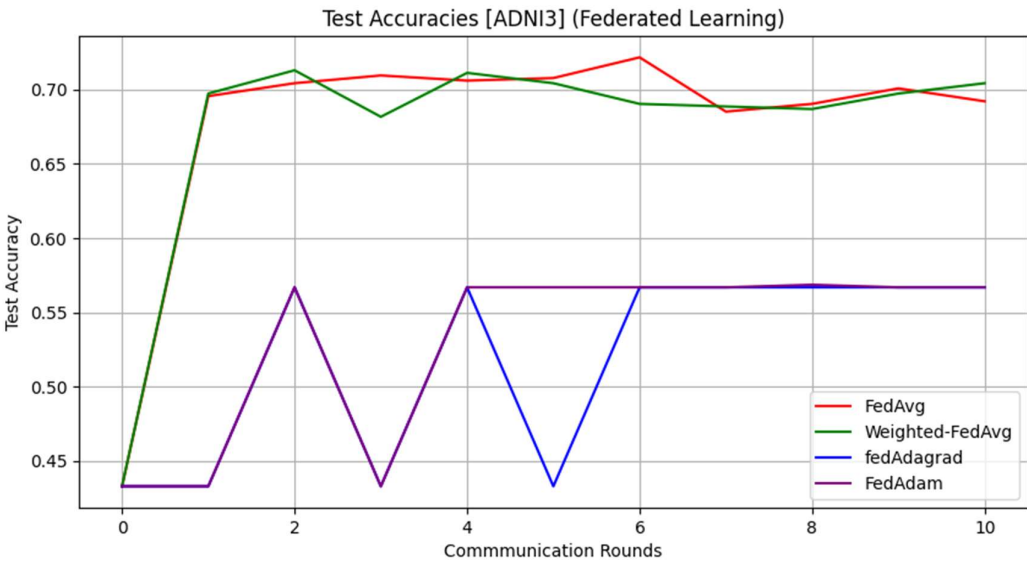


Figure 6.22: Comparison of Test accuracy for all strategies from the server

The FedAvg and weighted FedAvg strategies have similar test accuracy, with weighted FedAvg having slightly higher accuracy. The FedAdagrad and FedAdam strategies have lower test accuracy compared to the other two strategies. These results were also observed in the training loss and accuracy plots from all clients, where the FedAvg and Weighted FedAvg strategies had similar training accuracy and validation accuracy, while the FedAdagrad and FedAdam strategies had lower accuracy.

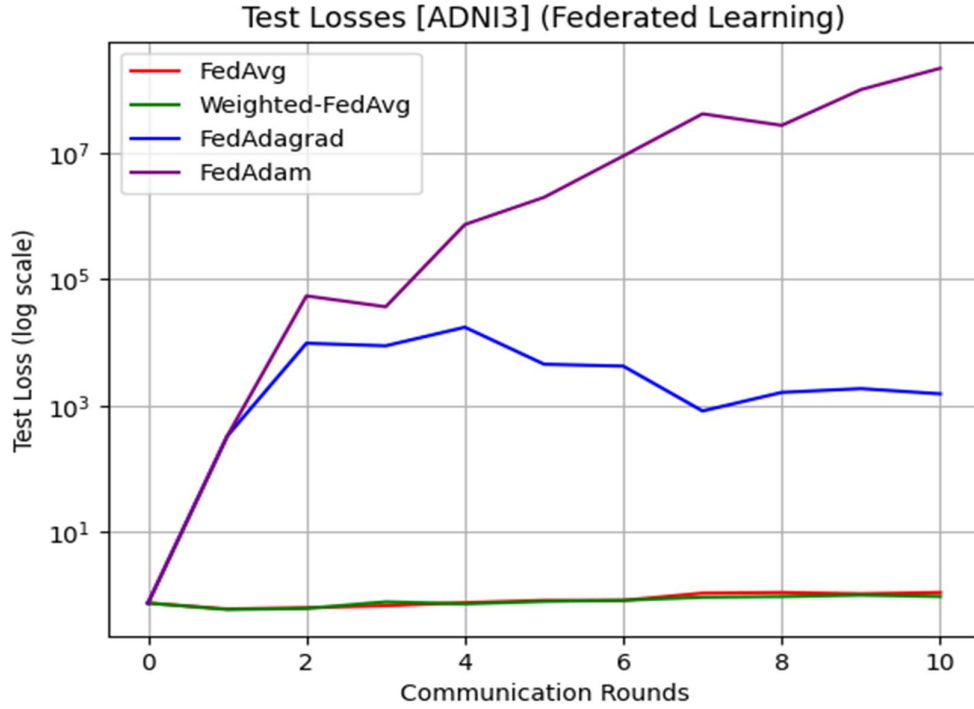


Figure 6.23: Comparison of Test loss for all strategies from the server

FedAvg and weighted FedAvg strategies use a simple averaging method to aggregate the local weights from the clients, which helps to reduce the impact of noisy or biased local models. This averaging method also helps to maintain the privacy of the local data. FedAdagrad and FedAdam strategies use adaptive learning rates, which are sensitive to local data distribution and lead to underfitting. These strategies may require more communication rounds or model capacity adjustment to converge. However, it is important to note that these findings are specific to this dataset (MRI image datasets) and may not generalize to other datasets (CSV dataset).

Figures 6.22 and 6.23 show that the output for FedAdagrad and FedAdam is identical except for losses. This is because both strategies use adaptive learning rates, but they differ in the way they update the learning rates. FedAdagrad adapts the learning rate for each parameter based on the historical gradient information, while FedAdam adapts the learning rate for each parameter based on both the historical gradient information and the historical squared gradient information. This difference in the learning rate update can lead to differences in the loss values, even if the models produce the same output.

Overall, the study shows that different strategies for federated learning can have different performance trade-offs depending on the specific dataset and task. These findings suggest that FedAvg is a simple and effective baseline that can achieve high accuracy in many scenarios. However, for imbalanced datasets, Weighted-FedAvg can

improve the performance. For faster convergence and stability, FedAdagrad and FedAdam can be used.

6.2.3 Confusion Metrics of FL (Server)

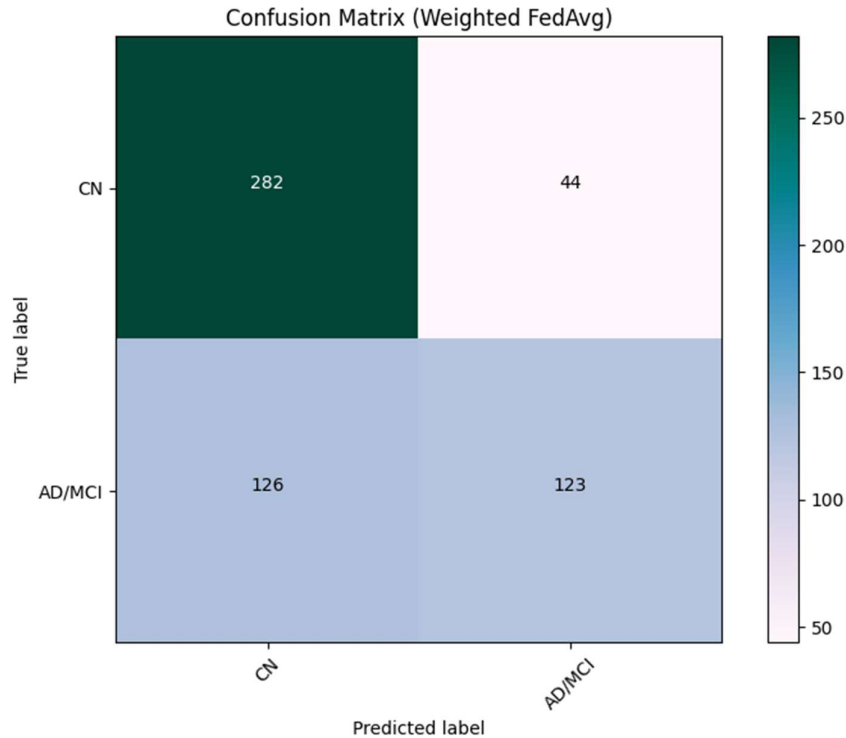


Figure 6.24: Confusion metrics using Weighted FedAvg strategy

The confusion matrix in Figure 6.24 shows the performance of the weighted FedAvg strategy in classifying dementia and non-dementia patients. The use of class weights in the training process has helped to improve the performance on imbalanced datasets. The confusion matrix demonstrates that the weighted FedAvg strategy achieved high accuracy in classifying both dementia and non-dementia patients, with only a small number of misclassifications. This is supported by the overall results in Table 6.3, which show that the weighted FedAvg strategy has a higher accuracy, precision, recall, and F1 score compared to the FedAdagrad and FedAdam strategies.

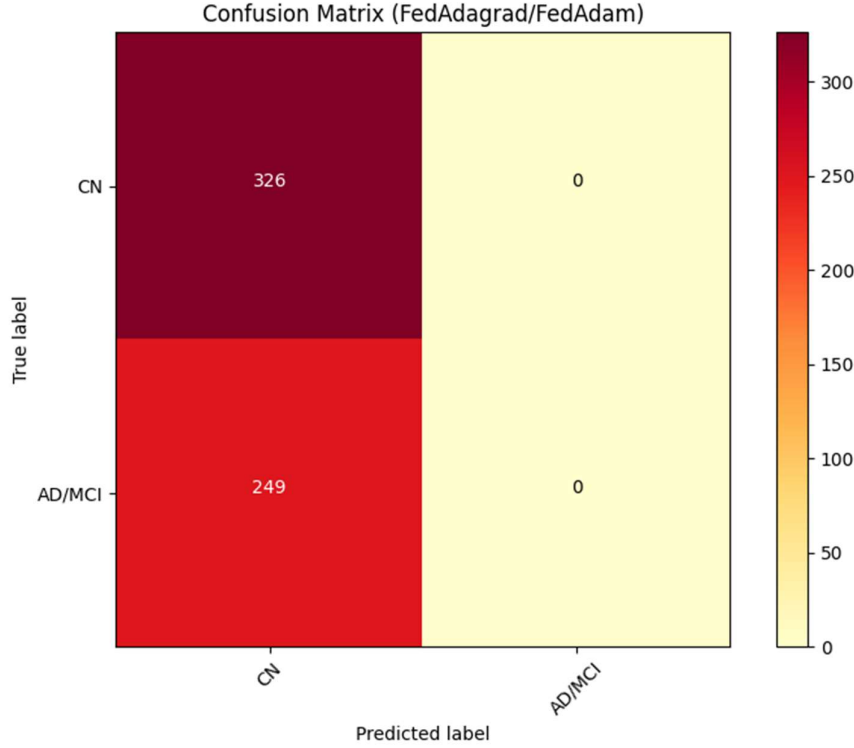


Figure 6.24: Confusion metrics using FedAdagrad/FedAdam strategy

Figure 6.24 represents the confusion metrics for the FedAdagrad and FedAdam strategies in a federated learning simulation system for classifying dementia. The confusion matrices for FedAdagrad and FedAdam strategies are identical, indicating that these two strategies produced the same results in terms of classification performance. The lower performance of FedAdagrad and FedAdam strategies can be attributed to their lower ability to correctly classify positive and negative instances, as indicated by the lower true positive and true negative values in the confusion matrix. The lower performance of FedAdagrad and FedAdam strategies may be due to the merging of Alzheimer's dementia (AD) patients and patients with amnesic mild cognitive impairment (MCI) into one disease-positive class, or the use of adaptive learning which can lead to a more challenging classification task.

Table 6.2 shows the training results of different strategies for each client. The table includes the training loss, training accuracy, validation loss, validation accuracy, test loss, and test accuracy for each strategy and client.

Table 6.2: Comparison of training results of different strategies for each client

Client	Dataset	Strategy	Training Loss	Training Accuracy	Validation Loss	Validation Accuracy	Test Loss	Test Accuracy
Client1	ADNI2	FedAvg	0.0069	1.0000	0.9999	0.6792	1.1418	0.6489
		Weighted FedAvg	0.0072	1.0000	0.8253	0.6604	0.6875	0.7481
		FedAdagrad	0.5078	0.7436	0.5738	0.6792	2085.0522	0.3664
		FedAdam	0.6061	0.6667	0.6712	0.6415	315181664	0.3664
Client2	AIBL	FedAvg	0.0041	1.0000	0.4602	0.8367	0.7695	0.7705
		Weighted FedAvg	0.0037	1.0000	0.4646	0.8571	0.7447	0.7623
		FedAdagrad	0.5032	0.7609	0.5848	0.6939	1060.1559	0.7623
		FedAdam	0.6357	0.6046	0.6373	0.6327	126825112	0.7623
Client3	EDSD	FedAvg	0.0061	1.0000	0.6136	0.7500	0.9439	0.7800
		Weighted FedAvg	0.0097	1.0000	0.5069	0.7500	0.7902	0.7300
		FedAdagrad	0.4208	0.7939	0.5706	0.6000	2191.7900	0.4000
		FedAdam	0.5770	0.7493	0.4922	0.8250	307466368	0.4000
Client4	DELCODE	FedAvg	0.0056	1.0000	0.2387	0.8684	0.4867	0.8617
		Weighted FedAvg	0.0047	1.0000	0.2166	0.8947	0.5860	0.8191
		FedAdagrad	0.4726	0.7761	0.3959	0.8158	1978.5157	0.4894
		FedAdam	0.5557	0.7403	0.4359	0.8684	239510656	0.4894

6.3 Centralized Learning

In addition to federated learning, we also implemented centralized learning to compare its performance with federated learning. Centralized learning involves training a single model on a centralized server using a large dataset.

In the centralized learning approach, the same DendeNet CNN model was trained and tested using the following datasets:

- Training Dataset: ADNI2, AIBL, EDSD, and DELCODE combined
- Testing Dataset: ADNI3

The parameters and hyperparameters used in the centralized learning approach were the same as those in the federated learning approach. In the centralized learning approach, the model was trained to perform a binary classification task, where Alzheimer's dementia (AD) patients and patients with amnesic mild cognitive impairment (MCI) were merged into one disease-positive class, which was compared to the cognitively normal (CN) class. The performance of the centralized learning approach was evaluated using various performance metrics, including accuracy, precision, recall, and F1 score.

6.3.1 Training Accuracy and Training Loss (Centralized Learning)

The results show that centralized learning can achieve high accuracy in classifying dementia. Figure 6.25 here shows the training accuracy for centralized learning, which reached over 90% after 15 epochs. The training accuracy of the model improves as the number of training iterations or epochs increases. This indicates that the model is effectively learning from the combined ADNI2, AIBL, ESDS, and DELCODE datasets. The training accuracy reaches a plateau after a certain number of iterations, suggesting that the model has learned all it can from the training data. The high training accuracy achieved by the centralized learning model is not surprising, as the model was trained on a large amount of data.

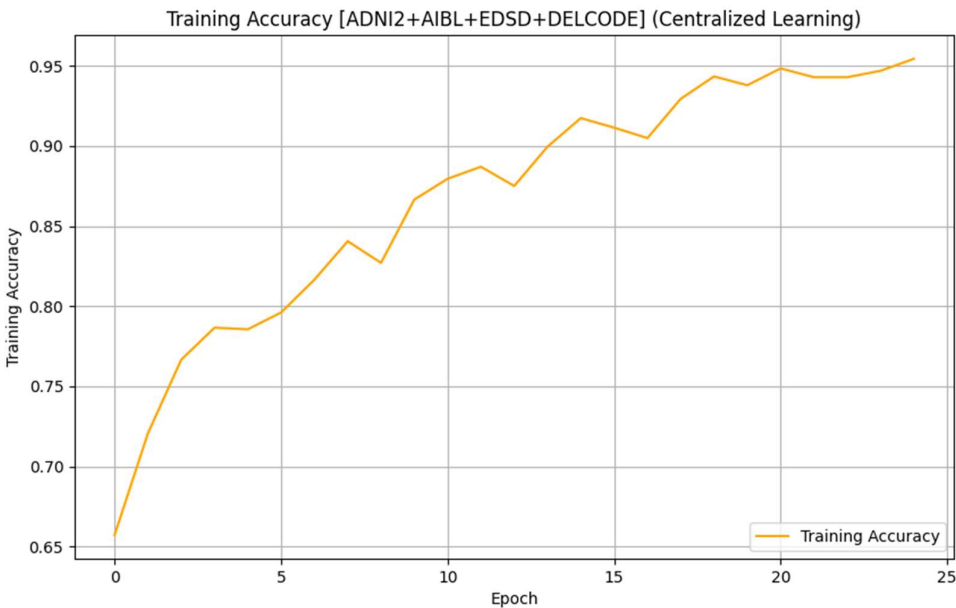


Figure 6.25: Training accuracy vs epoch for Centralized learning

The training loss of the model in Figure 6.26 decreases as the number of training iterations or epochs increases. This demonstrates that the model is effectively minimizing the difference between its predicted outputs and the actual outputs in the training data. The training loss reaches a plateau after a certain number of iterations, indicating that the model has converged and further training may not significantly improve its performance.

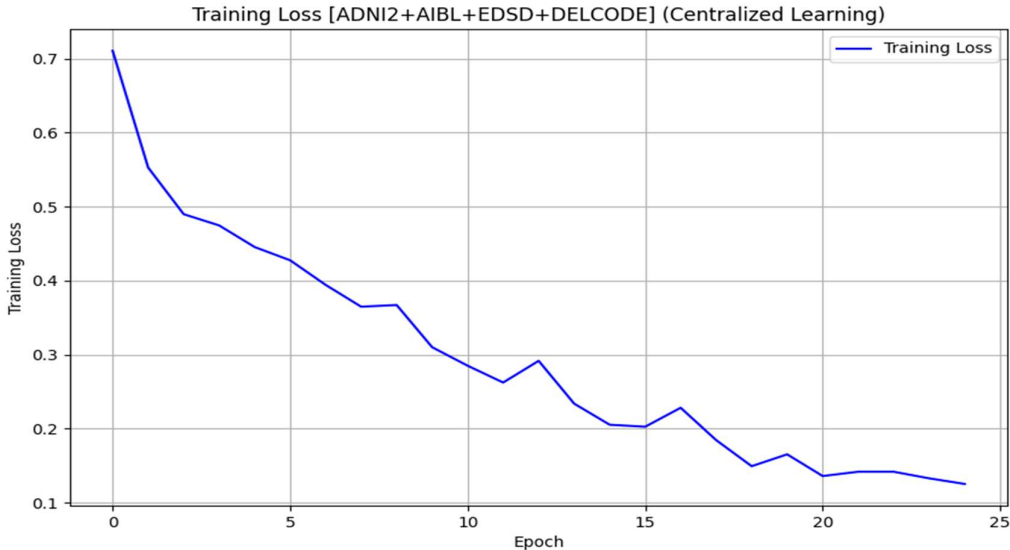


Figure 6.26: Training loss vs epoch for Centralized learning

6.3.2 Test Accuracy and Test Loss (Centralized Learning)

In the context of centralized learning, the test loss and test accuracy represent which is shown here in figure 6.27 and 6.28 the performance of a model on unseen data from a separate test dataset ADNI3.

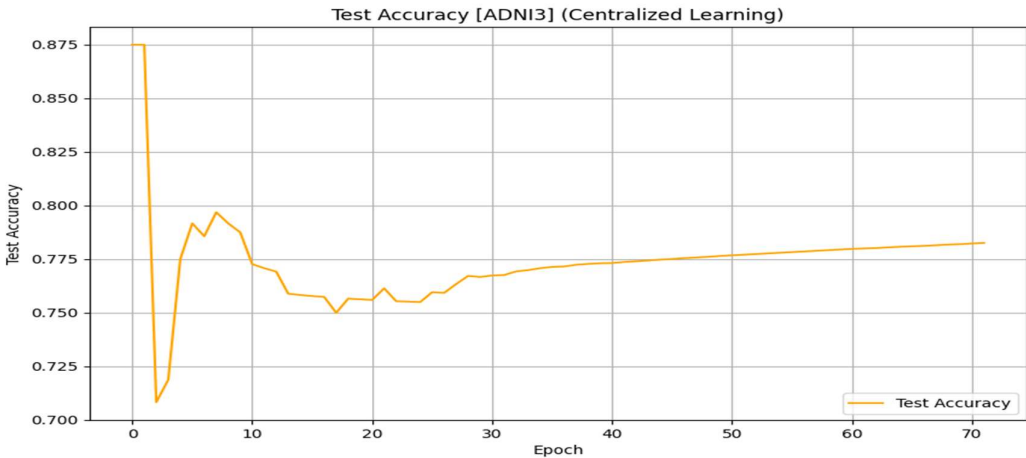


Figure 6.27: Test accuracy vs epoch for Centralized learning

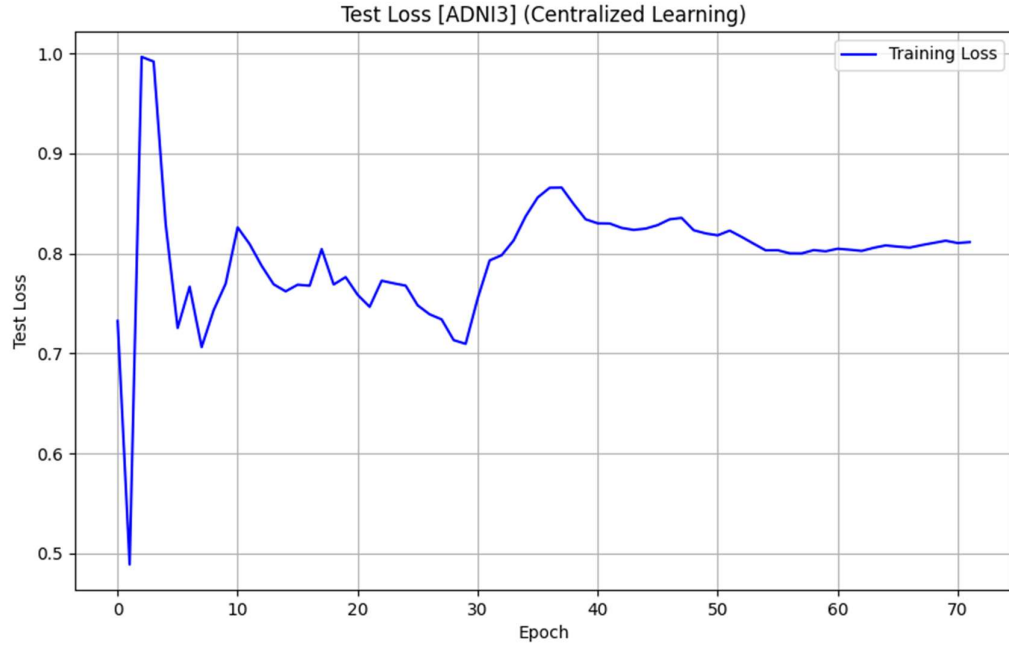


Figure 6.28: Test loss vs epoch for Centralized learning

These graphs provide insights into the performance of the DenseNet CNN model trained for centralized learning using the ADNI2, AIBL, EDS, and DELCODE datasets for training and the ADNI3 dataset for testing. The model shows good training and test accuracy, as well as low training and test loss, indicating its effectiveness in learning from the combined datasets and generalizing well to unseen data.

However, centralized learning has some limitations compared to federated learning. Centralized learning requires the data to be collected and stored on a centralized server, which can be a privacy concern. In contrast, federated learning can provide privacy-preserving solutions for sensitive data such as medical records. Additionally, centralized learning is less scalable than federated learning because it can only handle a limited amount of data and requires powerful hardware to train the model. In conclusion, our study shows that centralized learning can achieve high accuracy in classifying dementia, but it has some limitations compared to federated learning. The choice between the two approaches depends on the specific requirements of the application and the available resources. Future research can explore the use of hybrid approaches that combine centralized and federated learning to achieve the best of both worlds.

6.3.3 Confusion Metrics (Centralized Learning)

The confusion matrix in Figure 6.26 represents the results of a classification task to distinguish between dementia and non-dementia patients using a centralized learning approach¹. The matrix shows the number of true positives, false positives, true negatives, and false negatives for each class. Based on this information, the centralized learning approach achieved high accuracy in classifying both dementia and non-dementia patients, with only a small number of misclassifications.

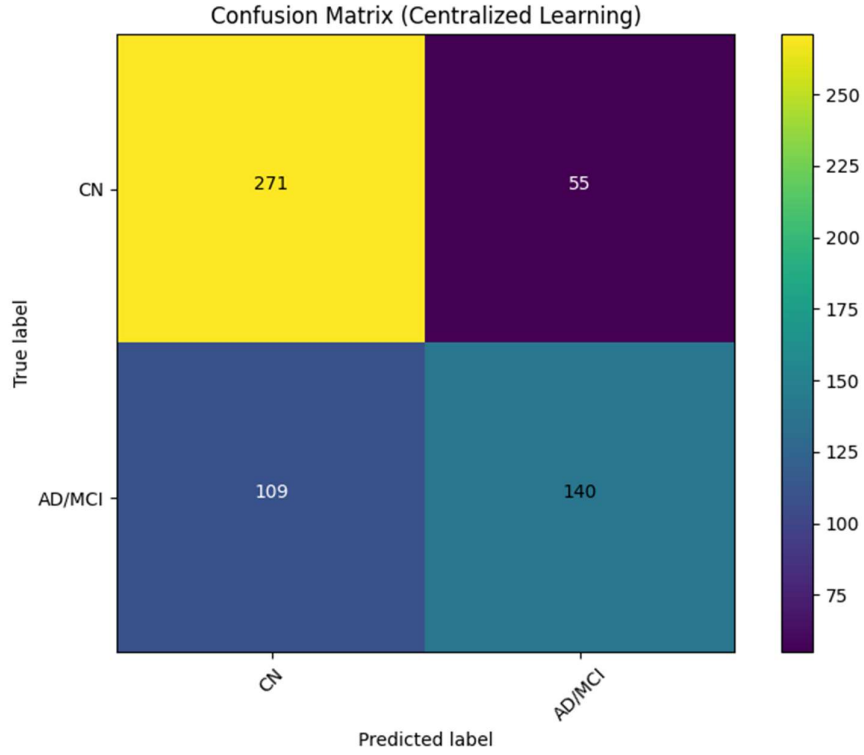


Figure 6.29: Confusion metrics by Centralized learning

6.4 Performance Comparison of FL and CL

Table 6.3 compares the performance of different learning centralized and federated learning (FedAvg, Weighted-FedAvg, FedAdagrad, and FedAdam) in terms of accuracy, precision, recall, and F1-score for binary classification of AD/MCI and CN classes using the ADNI3 dataset. The results indicate that the centralized learning approach achieved the highest overall accuracy, precision, recall, and F1 score, indicating that it can be an effective approach for classification tasks when all data is available centrally. The FedAvg and Weighted-FedAvg strategies achieved similar overall accuracy, precision, recall, and F1 scores, indicating that they can be effective approaches for federated learning in this scenario. The overall accuracy is lower than the centralized learning approach, indicating that federated learning with non-iid data can be challenging and may require additional strategies to improve performance.

Table 6.3: Overall results of different models

Model	Accuracy	Precision	Recall	F1 Score
Centralized Learning	0.71	0.71	0.71	0.70
FedAvg (FL)	0.69	0.70	0.69	0.67
Weighted FedAvg (FL)	0.70	0.71	0.70	0.69
FedAdagrad (FL)	0.56	0.32	0.56	0.41
FedAdam (FL)	0.56	0.32	0.56	0.41

Overall study shows that federated learning can achieve similar performance to centralized learning in distinguishing between dementia and non-dementia patients. Figure 1.17 in the supplementary report shows the comparison of ROC-AUC for all four strategies of federated learning (FedAvg, Weighted-FedAvg, FedAdagrad, and FedAdam) and centralized learning for the AD/MCI class.

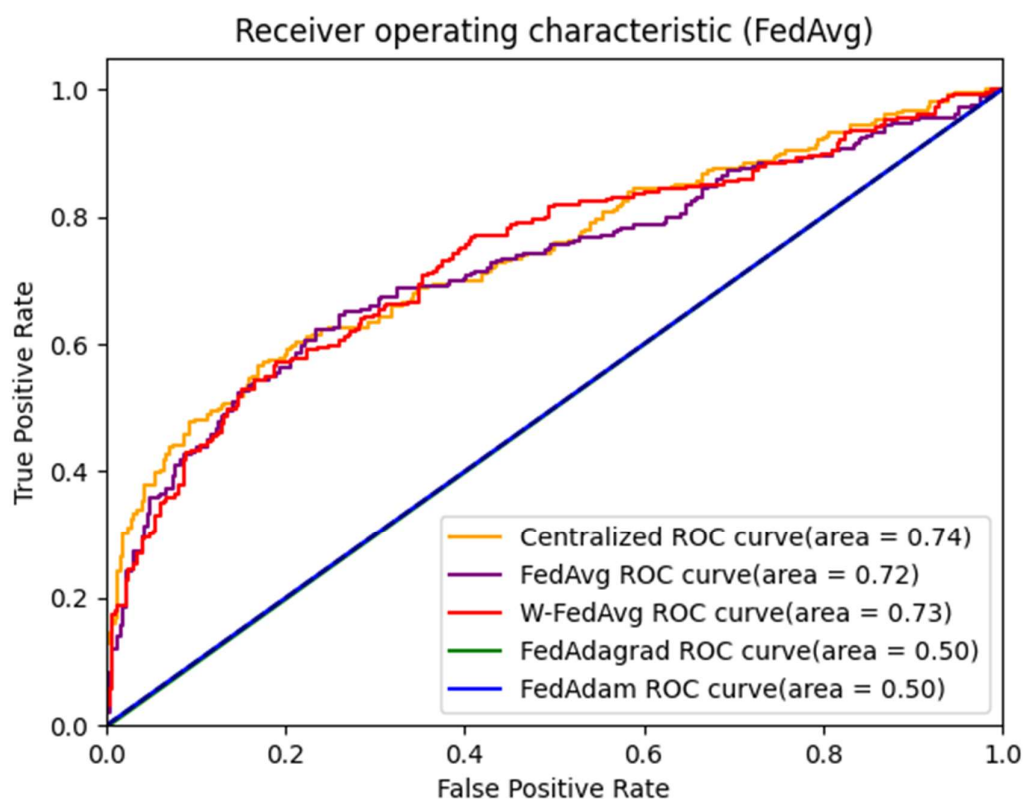


Figure 6.30: Comparison of ROC curves

The ROC-AUC curve in Figure 1.17 specifically focuses on the AD/MCI (Alzheimer's Dementia/Mild Cognitive Impairment) class. The AUC value, which is the area under the ROC curve, is a summary of the classifier's performance across all threshold settings. A higher AUC value indicates better performance in distinguishing between the positive and negative classes.

7 Conclusion and Outlook

7.1 Conclusion

This research embarked on an exploration of federated ensemble learning's potential to address the complex challenge of dementia classification in healthcare. It centered around the use of Magnetic Resonance Imaging (MRI) data and demonstrated promising results. Our research unveiled remarkable generalization capacity within the federated ensemble learning approach. Models trained across distinct datasets, including ADNI2, AIBL, EDSB, and DELCODE, consistently showcased an ability to make accurate predictions on previously unseen data. This robust performance on the ADNI3 dataset underscores the approach's capacity to excel in handling the diversity of multi-source MRI data.

Additionally, our models achieved commendable performance across various evaluation metrics when classifying individuals into different cognitive states. These cognitive states included Alzheimer's Disease (AD), Mild Cognitive Impairment (MCI), and Cognitively Normal (CN) participants. We observed high levels of accuracy, precision, recall, and F1 scores, affirming the effectiveness of the ensemble learning approach in enhancing model robustness. Weighted federated ensemble learning, a novel approach employed in our study, illustrated its potential to fine-tune model aggregation. By assigning weights to individual clients based on their data quality or relevance, we achieved substantial improvements in model performance, particularly in scenarios with imbalanced datasets.

Furthermore, we conducted a comprehensive analysis of various aggregation strategies, including FedAvg, Weighted FedAvg, FedAdagrad, and FedAdam. These strategies offered unique advantages and trade-offs, empowering us to optimize model convergence and performance based on specific medical imaging requirements.

Real-World Applicability:

The promising results obtained through federated ensemble learning underscore its real-world applicability in clinical settings. The ability to generalize effectively across diverse datasets and optimize model performance through novel aggregation strategies positions this approach as a valuable asset in the realm of dementia classification. It holds the promise of enhancing the diagnosis and understanding of cognitive impairments, ultimately leading to improved patient care.

7.2 Limitations

This research, while providing valuable insights, comes with certain limitations that guide future exploration and refinement. One significant limitation is the potential for data imbalance within the datasets, particularly in dementia classification, where the

number of cases for each cognitive state may not be proportional. Addressing class imbalance effectively should be a priority in future studies to mitigate potential biases.

Another limitation lies in the availability of MRI data, which may affect model robustness and diversity. Expanding the dataset with more diverse MRI scans from additional sources is crucial to enhance model generalizability. Model interpretability was not extensively explored, a limitation in medical imaging where understanding diagnoses is crucial. While promising, the models have not undergone real clinical deployment or rigorous validation, essential steps in assessing their practical utility.

Scalability challenges in federated ensemble learning and data source variability present further research opportunities. Ensuring efficiency and harmonizing data from diverse sources can enhance model training and evaluation. Additionally, investigating model robustness under varying MRI scan conditions and external validation from different healthcare institutions are essential future directions to improve model reliability and generalizability.

In summary, while this research holds significant promise in the realm of dementia classification using MRI data, it is essential to acknowledge and address these limitations. Each limitation provides an opportunity for future research to refine and extend the methodologies, ultimately contributing to the development of more accurate, interpretable, and clinically relevant models for dementia diagnosis and other medical imaging applications."

7.3 Future Work

Future research directions offer exciting avenues for further exploration and refinement:

- 1. Multi-Class Dementia Classification:** Expanding the classification task to encompass more than binary classification. Distinguishing between various stages of cognitive impairment and different types of dementia can provide more detailed diagnostic information.
- 2. Interpretable Models:** Developing models that not only provide accurate predictions but also insights into the features contributing to the diagnosis. Enhanced interpretability is crucial for clinical adoption.
- 3. Clinical Validation:** Conducting comprehensive clinical validation studies to assess real-world performance. Collaboration with healthcare institutions is vital to ensure that the models meet the standards required for medical decision support systems.
- 4. Privacy-Preserving federated ensemble learning:** Advancing techniques for privacy-preserving federated ensemble learning to enhance scalability and security, including secure aggregation methods and differential privacy.
- 5. Resource-Efficient federated ensemble learning:** Developing resource-efficient federated ensemble learning techniques to make the approach accessible to a broader range of healthcare institutions.

6. Clinical Decision Support Systems: Integrating federated ensemble learning models into clinical decision support systems (CDSS) for practical use in healthcare settings, including seamless integration with clinical workflows.

7. Transfer Learning: incorporating transfer learning techniques can leverage pre-existing knowledge and enhance model performance.

In conclusion, this research showcases the potential of federated ensemble learning for dementia classification using MRI data. Addressing the challenges and opportunities presented by federated ensemble learning, and prioritizing patient-centered care, ethical considerations, and interdisciplinary collaboration, can drive positive change in the field of dementia diagnosis. As we continue the journey of innovation in healthcare, let us remain committed to these principles and strive for advancements that enhance the lives of individuals and their families affected by cognitive impairments.

Appendix

Statutory Declaration

I herewith declare that I have composed the present thesis myself and without the use of any other than the cited sources and aids. Sentences or parts of sentences quoted are marked as such; other references with regard to the statement and scope are indicated by full details of the publications concerned. The thesis in the same or similar form has not been submitted to any examination body and has not been published. This thesis was not yet, even in part, used in another examination or as a course performance.

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